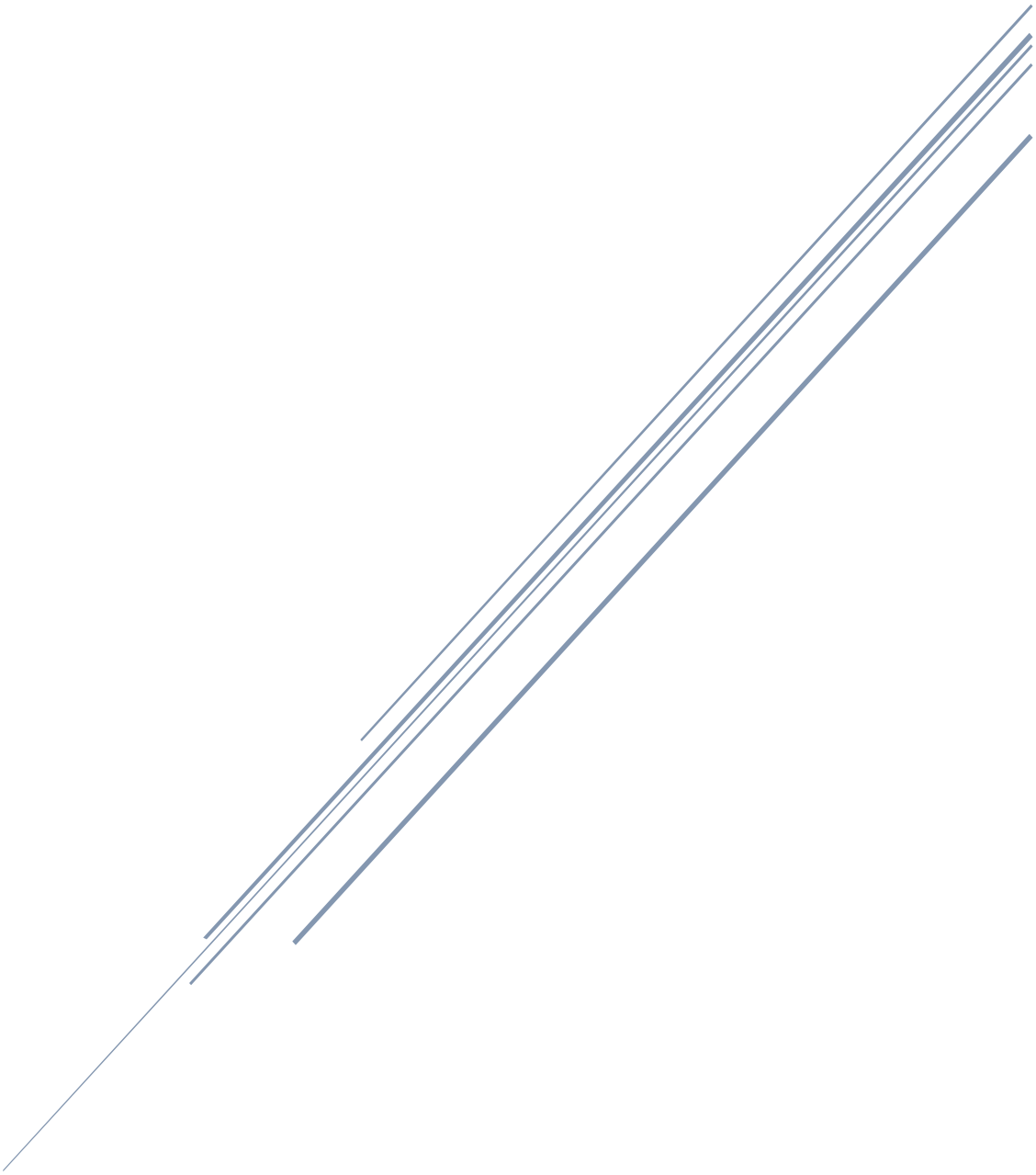
HOUSE PRICE PREDICTION



Meenu G

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Context

A house value is simply more than location and square footage. Like the features that make up a person, an educated party would want to know all aspects that give a house its value. For example, you want to sell a house and you don't know the price which you may expect — it can't be too low or too high. To find house price you usually try to find similar properties in your neighbourhood and based on gathered data you will try to assess your house price.

Objective

A dataset pertaining to the year 2014-2015 for each house sold/bought in certain regions of USA is given for the price analysis. This data is having property level detail such as price, size, location and so on. The main objective of the current task is to perform Exploratory Data Analysis (EDA) — to understand the distribution of variables, spot patterns, identify outliers, and explore relationships between features before moving on to any advanced modelling.

Data Dictionary

1. cid: a notation for a house
2. dayhours: Date house was sold
3. price: Price is prediction target
4. room_bed: Number of Bedrooms/House
5. room_bath: Number of bathrooms/bedrooms
6. living_measure: square footage of the home
7. lot_measure: square footage of the lot
8. ceil: Total floors (levels) in house
9. coast: House which has a view to a waterfront
10. sight: Has been viewed
11. condition: How good the condition is (Overall)
12. quality: grade given to the housing unit, based on grading system
13. ceil_measure: square footage of house apart from basement
14. basement_measure: square footage of the basement
15. yr_built: Built Year
16. yr_renovated: Year when house was renovated
17. zipcode: zip
18. lat: Latitude coordinate
19. long: Longitude coordinate
20. living_measure15: Living room area in 2015(implies-- some renovations) This might or might not have affected the lotsize area
21. lot_measure15: lotSize area in 2015(implies-- some renovations)
22. furnished: Based on the quality of room
23. total_area: Measure of both living and lot

Data Overview

Following tables showing the first few and last few rows of dataset.

	cid	dayhours	price	room_bed	room_bath	living_measure	lot_measure	ceil	coast	sight	...	basement	yr_built	yr_renovated	zipcode	lat
0	3876100940	20150427T000000	600000	4.0	1.75	3050.0	9440.0	1	0	0.0	...	1250.0	1966	0	98034	47.7228
1	3145600250	20150317T000000	190000	2.0	1.00	670.0	3101.0	1	0	0.0	...	0.0	1948	0	98118	47.5546
2	7129303070	20140820T000000	735000	4.0	2.75	3040.0	2415.0	2	1	4.0	...	0.0	1966	0	98118	47.5188
3	7338220280	20141010T000000	257000	3.0	2.50	1740.0	3721.0	2	0	0.0	...	0.0	2009	0	98002	47.3363
4	7950300670	20150218T000000	450000	2.0	1.00	1120.0	4590.0	1	0	0.0	...	0.0	1924	0	98118	47.5663

Table 1: First few rows of dataset

	cid	dayhours	price	room_bed	room_bath	living_measure	lot_measure	ceil	coast	sight	...	basement	yr_built	yr_renovated	zipcode	
21608	203600600	20150310T000000	685530	4.0	2.50	3130.0	60467.0	2	0	0.0	...	0.0	1996	0	98014	4
21609	625049281	20140521T000000	535000	2.0	1.00	1030.0	4841.0	1	0	0.0	...	110.0	1939	0	98103	4
21610	424069018	20140905T000000	998000	3.0	3.75	3710.0	34412.0	2	0	0.0	...	800.0	1978	0	98075	4
21611	7258200055	20150206T000000	262000	4.0	2.50	1560.0	7800.0	2	0	0.0	...	0.0	1997	0	98168	4
21612	8805900430	20141229T000000	1150000	4.0	2.50	1940.0	4875.0	2	0	0.0	...	0.0	1925	0	98112	4

Table 2:Last five rows of dataset

Shape of dataset: Data contains 21613 rows and 23 columns

Data types: There are total 7 categorical variables which are shown as object data types and 16 numerical variables which has data type as float64 or int64.

```
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   cid              21613 non-null  int64
 1   dayhours         21613 non-null  object
 2   price            21613 non-null  int64
 3   room_bed         21505 non-null  float64
 4   room_bath        21505 non-null  float64
 5   living_measure   21596 non-null  float64
 6   lot_measure      21571 non-null  float64
 7   ceil             21571 non-null  object
 8   coast            21612 non-null  object
 9   sight            21556 non-null  float64
 10  condition        21556 non-null  object
 11  quality          21612 non-null  float64
 12  ceil_measure     21612 non-null  float64
 13  basement         21612 non-null  float64
 14  yr_built         21612 non-null  object
 15  yr_renovated     21613 non-null  int64
 16  zipcode          21613 non-null  int64
 17  lat              21613 non-null  float64
 18  long             21613 non-null  object
 19  living_measure15 21447 non-null  float64
 20  lot_measure15    21584 non-null  float64
 21  furnished        21584 non-null  float64
 22  total_area       21584 non-null  object
```

Table 3: Data types of variables

Some of these variables like long, total_area are likely to be numerical data types that might need to be converted for further processing

Unique values in data set: Following table is listing the number of unique values for each variable

cid	21436
dayhours	372
price	3625
room_bed	13
room_bath	30
living_measure	1038
lot_measure	9765
ceil	7
coast	3
sight	5
condition	6
quality	12
ceil_measure	946
basement	306
yr_built	117
yr_renovated	70
zipcode	70
lat	5034
long	753
living_measure15	774
lot_measure15	8682
furnished	2
total_area	11145
dtype: int64	

Table 4:Unique values in data set

Checking for missing values: Certain columns have missing data which are to be treated in upcoming sections

cid	0
dayhours	0
price	0
room_bed	108
room_bath	108
living_measure	17
lot_measure	42
ceil	42
coast	1
sight	57
condition	57
quality	1
ceil_measure	1
basement	1
yr_built	1
yr_renovated	0
zipcode	0
lat	0
long	0
living_measure15	166
lot_measure15	29
furnished	29
total_area	29
dtype: int64	

Table 5:Missing values per feature

Checking for duplicate values: There are no duplicate values in data

Missing value Treatment

In here we are dropping all the rows that has missing values or null values. This has resulted in a marginal reduction of the number of rows – only 1.34% reduction. Total rows reduced from 21613 to to 21323.

Columns like total_area, long, condition also had invalid entries which appeared as '$', such rows were also dropped.

Outlier treatment

The dataset contains multiple outliers across various features. These outliers remain unchanged and have not been removed or modified in the current analysis. They are visible in the upcoming univariate plots.

Univariate Analysis

Statistical summary:

	cid	price	room_bed	room_bath	living_measure	lot_measure	sight	quality	ceil_measure	basement	yr_renovated	zipcode	lat	living_measure15	lot_measure15	furnished
count	2.138700e+04	2.138700e+04	21387.000000	21387.000000	21387.000000	2.138700e+04	21387.000000	21387.000000	21387.000000	21387.000000	21387.000000	21387.000000	21387.000000	21387.000000	21387.000000	21387.000000
mean	4.577761e+09	5.403520e+05	3.370880	2.114941	2080.473278	1.511142e+04	0.234862	7.657923	1789.025670	291.447608	83.892037	98077.891757	47.559999	1987.044420	12762.132557	0.196942
std	2.877364e+09	3.681089e+05	0.930488	0.769806	918.942968	4.144908e+04	0.767248	1.176458	828.581674	442.684460	400.511148	53.498113	0.138583	685.763628	27241.164726	0.397697
min	1.000102e+06	7.500000e+04	0.000000	0.000000	290.000000	5.200000e+02	0.000000	1.000000	290.000000	0.000000	0.000000	98001.000000	47.155900	399.000000	651.000000	0.000000
25%	2.122054e+09	3.210000e+05	3.000000	1.750000	1430.000000	5.040000e+03	0.000000	7.000000	1190.000000	0.000000	0.000000	98033.000000	47.470650	1490.000000	5100.000000	0.000000
50%	3.904920e+09	4.500000e+05	3.000000	2.250000	1910.000000	7.620000e+03	0.000000	7.000000	1560.000000	0.000000	0.000000	98065.000000	47.571700	1840.000000	7620.000000	0.000000
75%	7.307250e+09	6.450000e+05	4.000000	2.500000	2550.000000	1.068750e+04	0.000000	8.000000	2210.000000	560.000000	0.000000	98117.000000	47.678000	2360.000000	10085.000000	0.000000
max	9.900000e+09	7.700000e+06	33.000000	8.000000	13540.000000	1.651359e+06	4.000000	13.000000	9410.000000	4820.000000	2015.000000	98199.000000	47.777600	6210.000000	871200.000000	1.000000

Table 6: A glimpse of Statistical summary of dataset

Price: Univariate analysis

The histogram is right skewed with lot of outliers beyond the upper whisker.

Mean: $540352.04
Median: $450000.00
Mode: $350000.00
Percentage of Outliers: 5.28%
Percentage of Inliers: 94.72%

Almost 50 to 60% houses are in the price range of $0.2M to $0.5M.

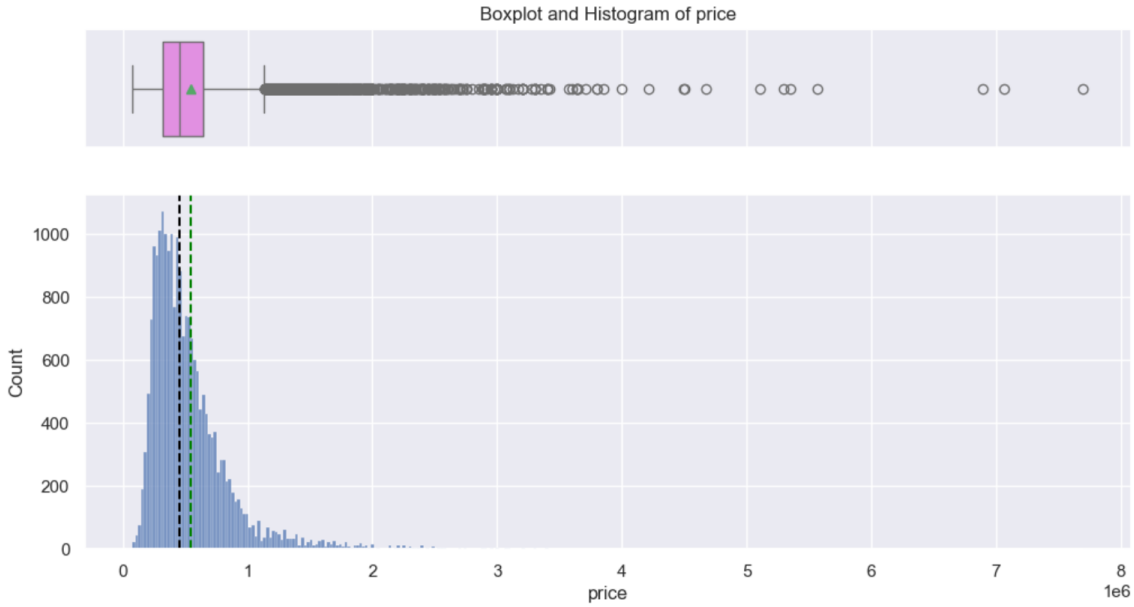


Figure 1:Boxplot and histogram of price along with mean and median as vertical lines

Room_bed: Univariate Analysis

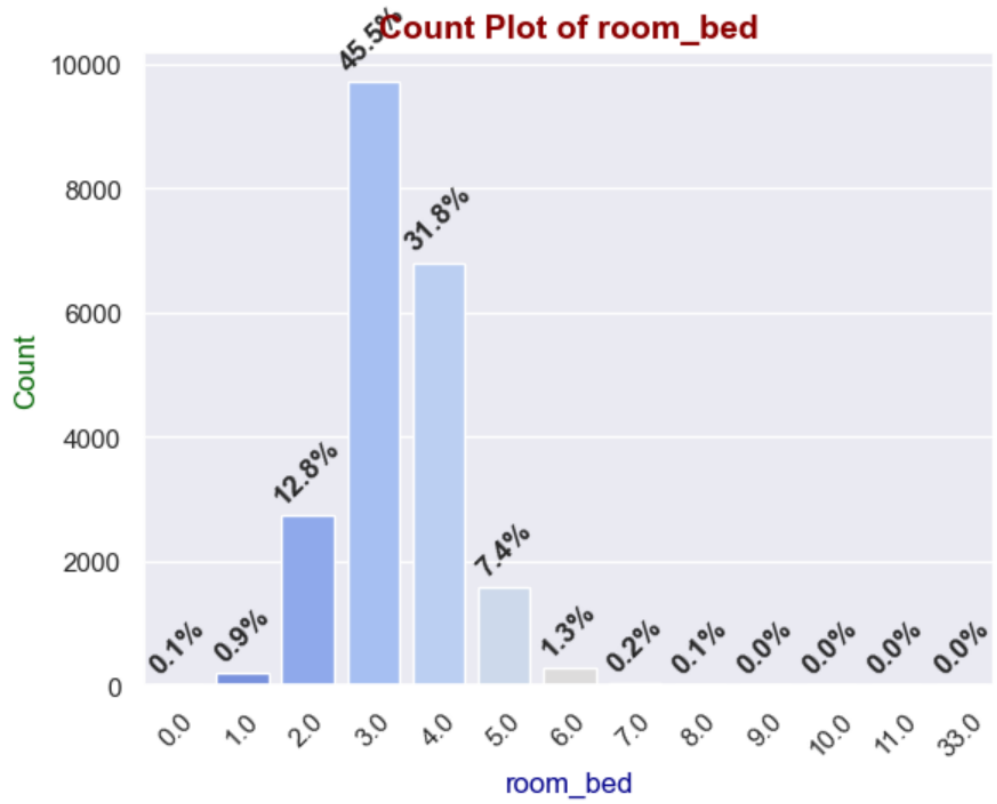


Figure 2

Three-bedroom homes are the most in demand, followed by properties with four, two, and five bedrooms. The number of bedrooms ranges from a minimum of zero to a maximum of 33. Zero bedrooms indicate probably a studio apartment or open living space and 33 bed room houses can be a mansion or hotel etc

Room_bath-Univariate Analysis

The room_bath variable captures the ratio of bathrooms to bedrooms. Since it's a ratio, the values are typically non-integer. The distribution shows that the most common ratio is 2.5, followed by 1.0 and 1.75. Over 97% of the data points fall between range of 1.0 to 3.5.

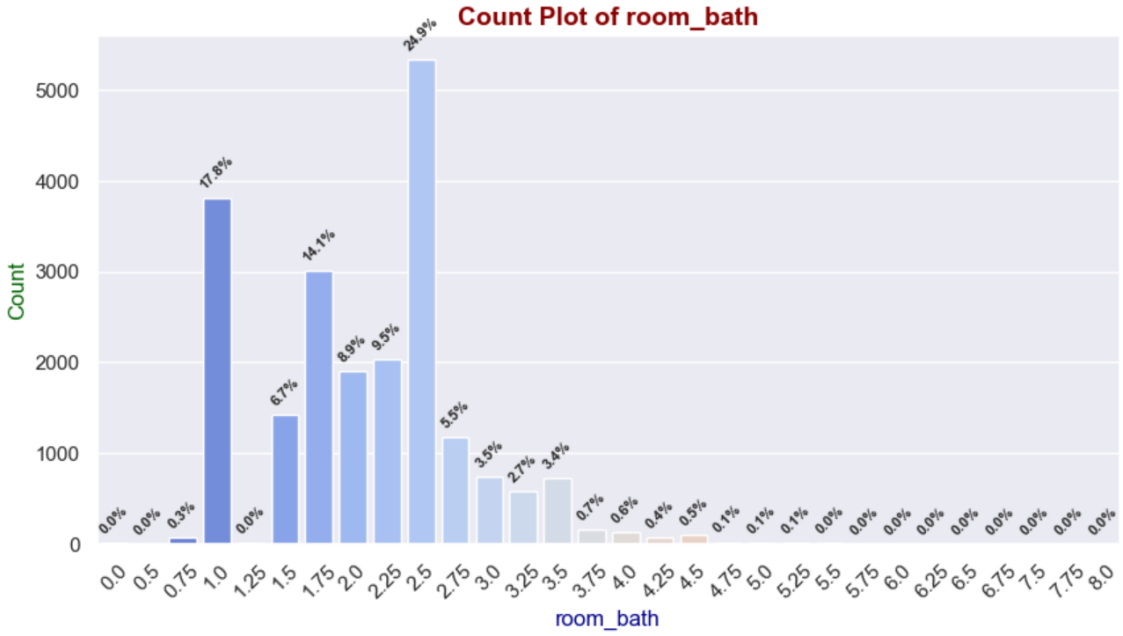


Figure 3

count	21387.000000
mean	2.114941
std	0.769806
min	0.000000
25%	1.750000
50%	2.250000
75%	2.500000
max	8.000000

Table 7:Statistical summary of the room_bath

- The most common bathroom-to-bedroom ratio is 2.5, meaning these properties tend to have more than double the number of bathrooms compared to bedrooms. This pattern often points to hotel-style buildings or luxury accommodations.
- A ratio of 1.0, seen in about 17.8% of properties, suggests a balanced setup—one bathroom per bedroom, typical of standard family homes.
- The lowest ratio is 0, indicating properties that may not have any bathrooms—possibly very basic units, utility spaces, or missing data.
- On the other end, the maximum ratio is 8, meaning some properties have eight times as many bathrooms as bedrooms. These are likely high-end estates, guest lodges, or unusual outliers worth a second look.

Living_measure -Univariate Analysis

This measure represents the total interior living space of a home. Over 60% of properties fall within the 1,000 to 2,500 square feet range. The distribution is right-skewed, indicating a concentration of mid-sized homes along with a few significantly larger properties. Overall, home sizes span a wide range—from as small as 290 sq ft to a maximum of 13,450 sq ft.

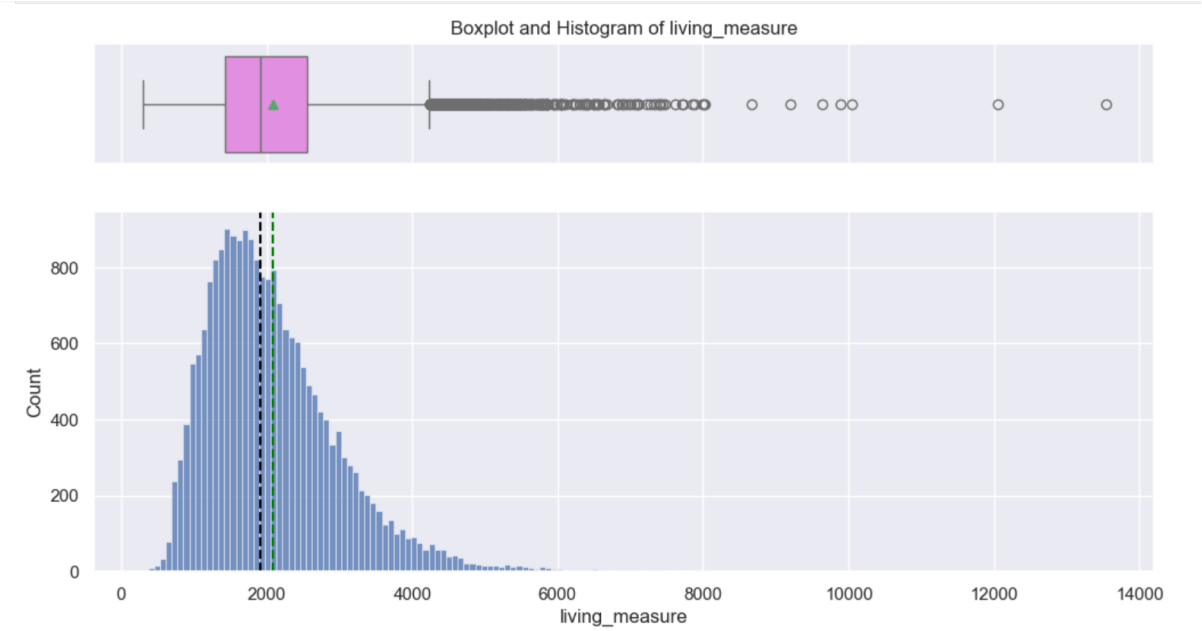


Figure 4

count	21387.000000
mean	2080.473278
std	918.942968
min	290.000000
25%	1430.000000
50%	1910.000000
75%	2550.000000
max	13540.000000

Table 8:Staistical Summary of living_measure variable

Lot_measure -Univariate Analysis

The box plot reveals a significant number of outliers, suggesting that many large plots have been sold. Compared to the living_measure, this feature has a higher proportion of outliers—approximately 11%. When outliers are excluded, the histogram shows that nearly 86% of lot sizes fall between 4,000 and 11,000 square feet.

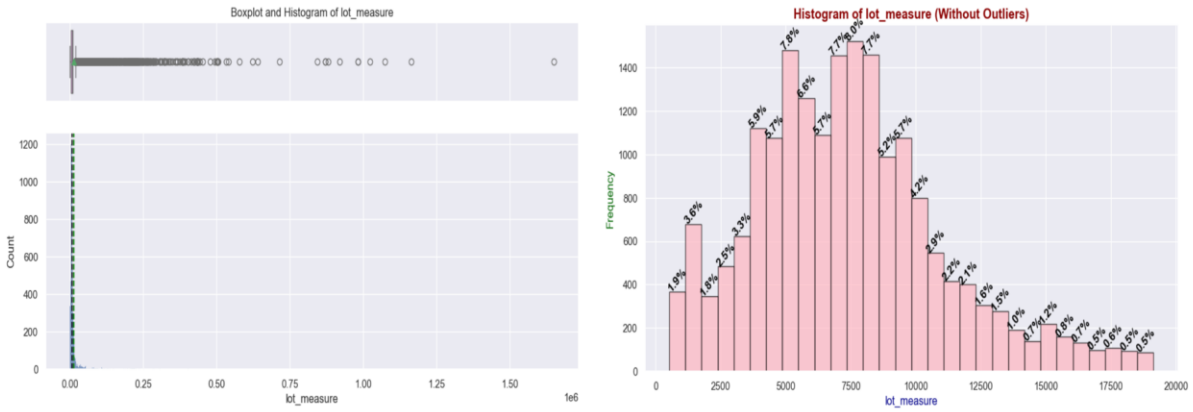


Figure 5:Boxplot and histogram of lot_measure variable on left side and histogram of lot_measure without outliers on right side

sight -Univariate Analysis

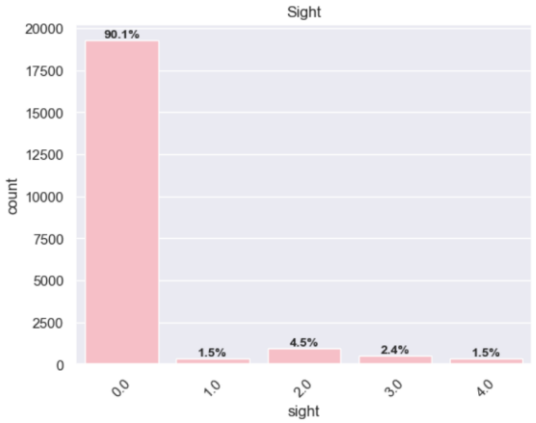


Figure 6:Count plot of sight variable with percentage count for each value of sight variable

90% of the properties didn't have any sightings and some properties have sightings as high as 4. This shows that the sold properties didn't have a sighting as such.

Quality -Univariate Analysis

Quality ratings range from 0 to 13, with the majority of properties scoring 5 or higher. The most frequently assigned rating is 7, followed by 8 and 9.

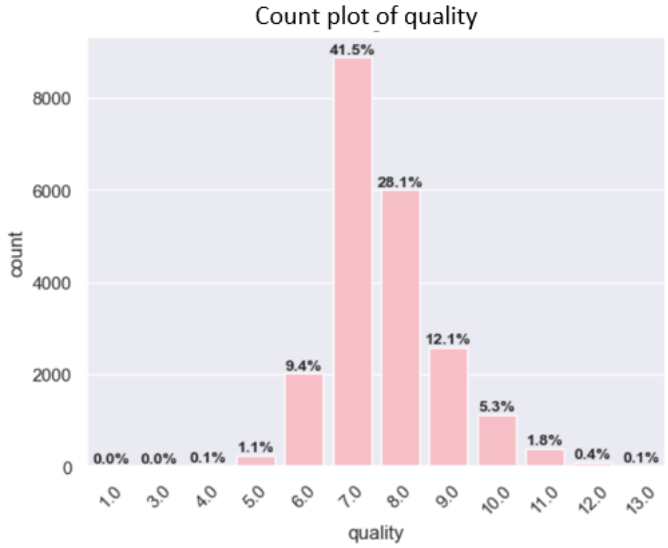


Figure 7:Count plot of quality

Ceil: Univariate Analysis

The variable ceil represents the number of floors or levels in a house. Most houses have just one floor, followed by two floors, and then one-and-a-half floors. Houses with more than three floors are also present in the data, though in smaller numbers compared to the others.

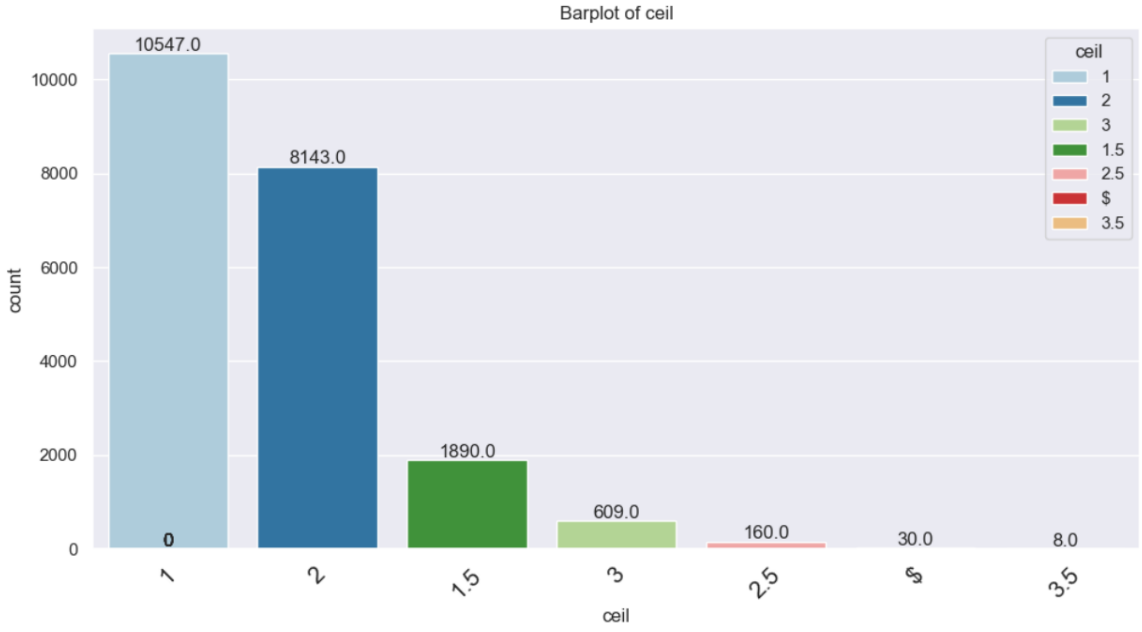


Figure 8

Ceil_measure – Univariate Analysis

The ceil_measure variable is right-skewed, with values ranging from 290 to 9410. Most values are clustered toward the lower end, and only a few properties have very high ceiling measurements. Interestingly, the minimum and mode are quite similar to those of the living_measure variable, hinting at a possible relationship between the two. Around 65% of the values fall between 1000 and 2000, with a noticeable concentration near 1300. This suggests that most ceilings follow a common standard, while the longer right tail reflects a few larger or more luxurious homes.

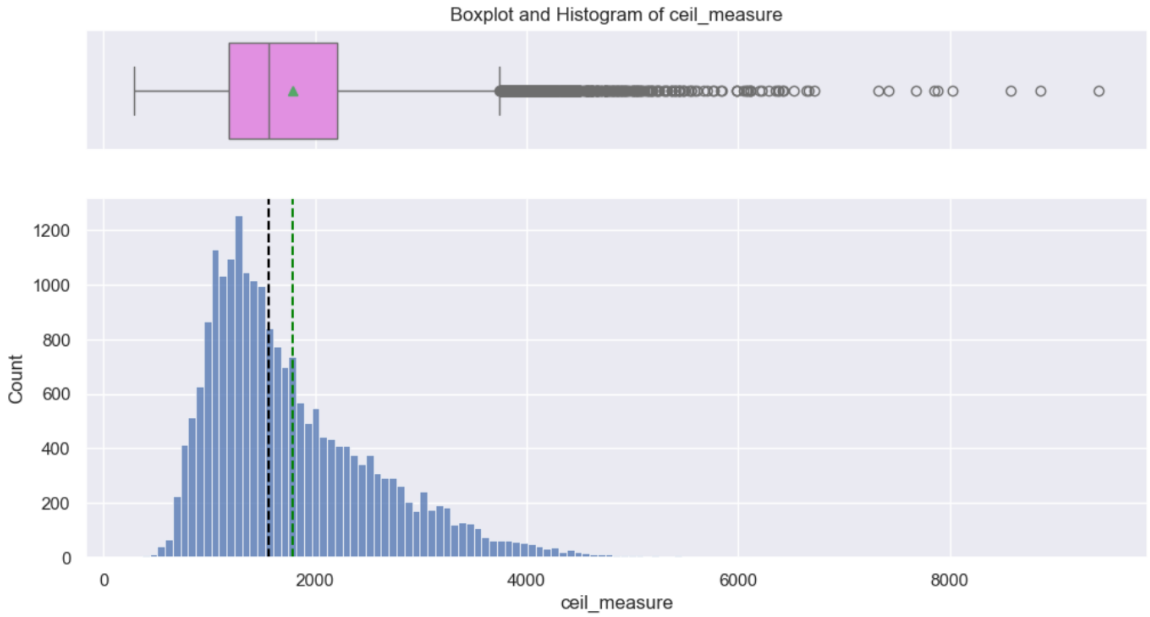


Figure 9

Basement: Univariate analysis

The basement_measure variable is heavily right-skewed, with a large number of properties — about 62% — having no basement at all, indicated by a measure of 0. This results in a sharp peak at 0 on the histogram. However, some properties do have basements, with sizes going up to 4820, creating a long right tail and several outliers. The distribution clearly reflects that while basements are uncommon, the ones that do exist vary widely in size.

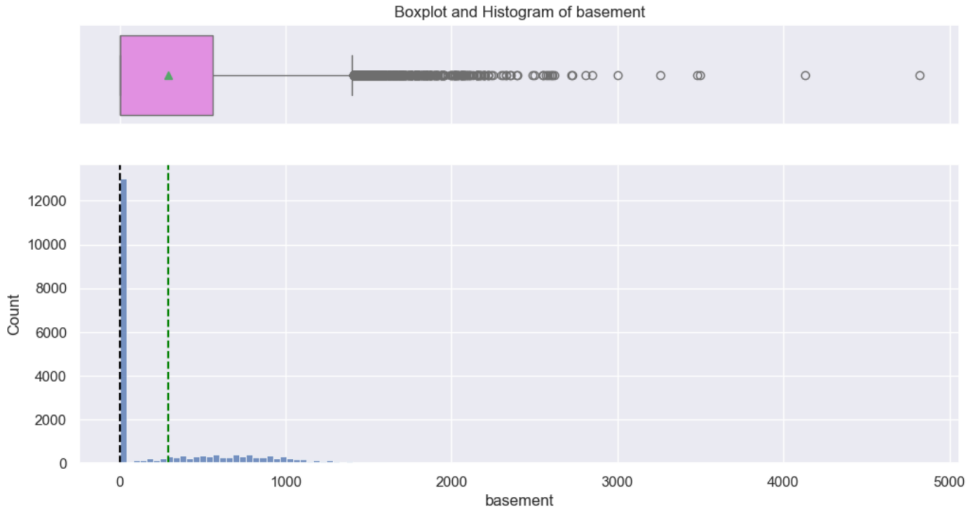


Figure 10

Yr_renovated:Univariate analysis

About 95% of the properties were never renovated. Among the remaining 5% that were renovated, 2% were updated after the year 2000, while the other 2% were renovated before 2000.

	years	count	percentage
0	Not-Renovated	20488	95.796512
1	2000+	450	2.104082
2	1950-1999	439	2.052649
3	1900-1949	10	0.046757

Table 9

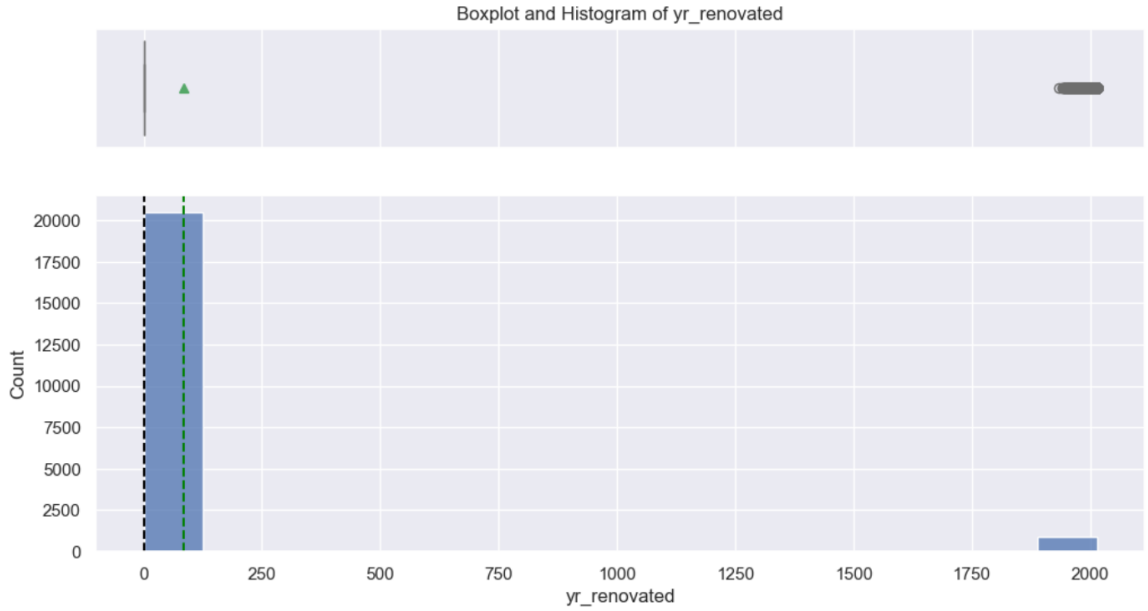


Figure 11:Year renovated bar plot

Living_measure15: Univariate Analysis

This measurement reflects the state of the property as of 2015 and can help identify if a house has been renovated — especially when it's different from the living area size.

The histogram and boxplot show many high-end outliers, which means the data is right-skewed. Its shape is quite similar to the living area distribution, but there's a noticeable peak around 1540 sq ft, with fewer houses in the nearby size ranges. This suggests that renovations have often increased the property's size to around that value.

Also, the minimum size has gone up from 289 sq ft (in the original living area) to 399 sq ft, and the maximum has dropped from 13,540 sq ft to 6,210 sq ft — possibly indicating that some renovations actually reduced the total livable area.

```
Analysis for 'living_measure15':
Mean: 1987.04
Median: 1840.00
Mode: 1540.00
Percentage of Outliers: 2.52%
Percentage of Inliers: 97.48%
```

```
count    21387.000000
mean      1987.044420
std        685.763628
min        399.000000
25%       1490.000000
50%       1840.000000
75%       2360.000000
max       6210.000000
Name: living_measure15, dtype: float64
```

Table 10:Statistical summary of living measure

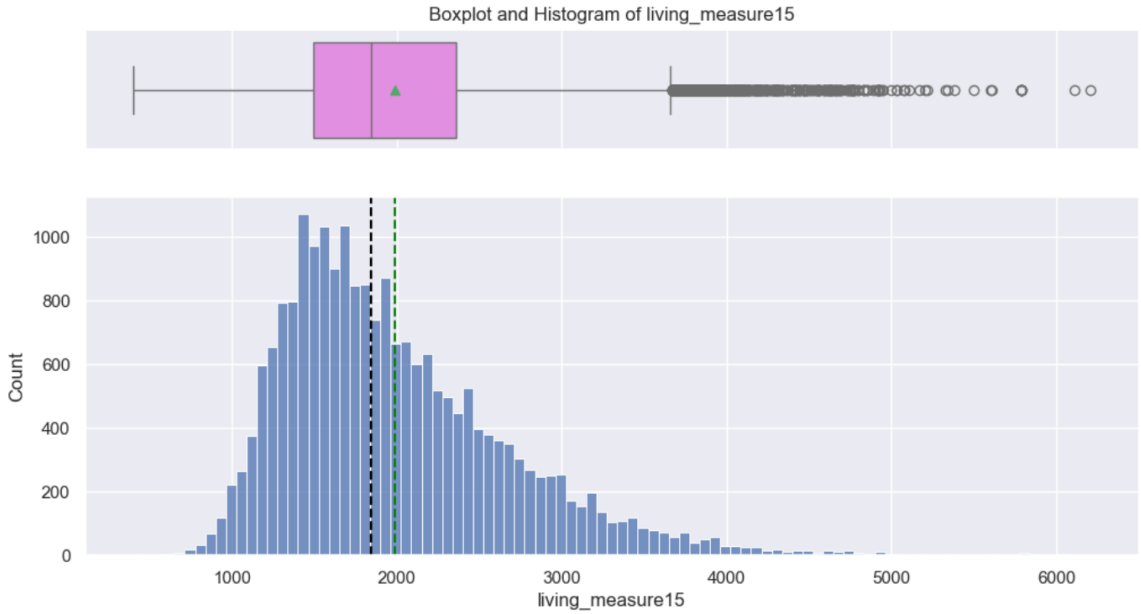


Figure 12

Lot_measure15: Univariate Analysis

```
count     21387.000000
mean      12762.132557
std       27241.164726
min         651.000000
25%        5100.000000
50%        7620.000000
75%       10085.000000
max      871200.000000
Name: lot_measure15, dtype: float64
```

```
Analysis for 'lot_measure15':
Mean: 12762.13
Median: 7620.00
Mode: 5000.00
Percentage of Outliers: 10.17%
Percentage of Inliers: 89.83%
```

Table 11: Statistical summary for lot_measure15

Similar to lot_meausre variable, there are multiple outliers and histogram is right skewed. The smallest lot size went up from 520 sq ft to 651 sq ft, and the largest lot size dropped from 1.6 million to 870,000 sq ft. This likely means some of the bigger properties were split into smaller ones during

renovations. Even with these changes, the most common lot size in 2015 stayed the same at 5,000 sq ft.

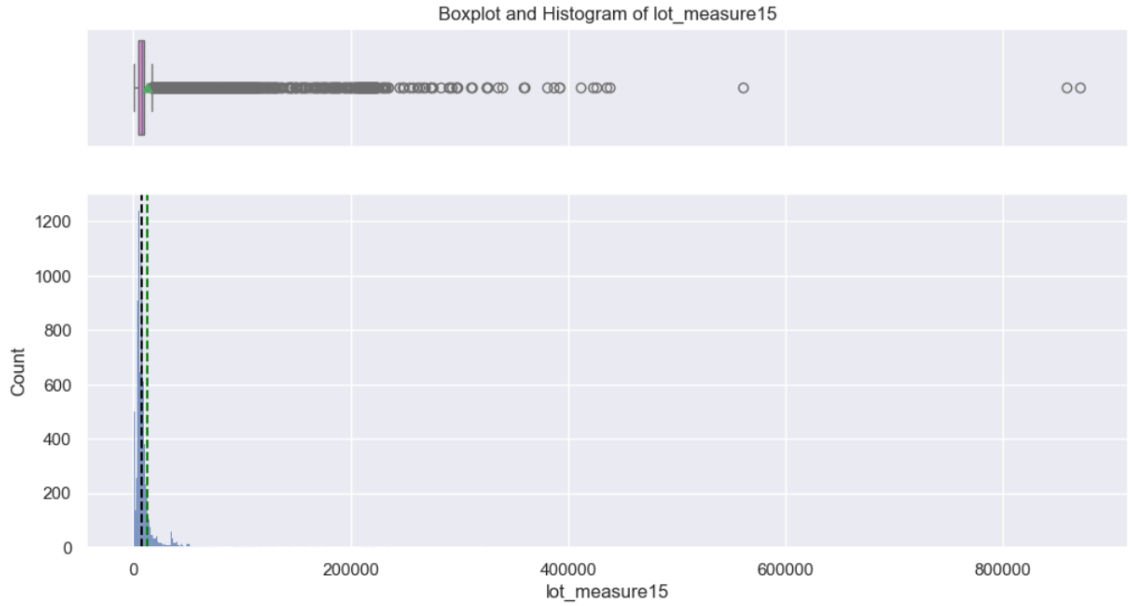


Figure 13

Furnished: Univariate Analysis

This is a binary variable says whether the houses are furnished or not. 80% of the houses are not furnished which takes value '0' and only 20% of the houses are furnished which takes value '1' as shown in the pie chart.

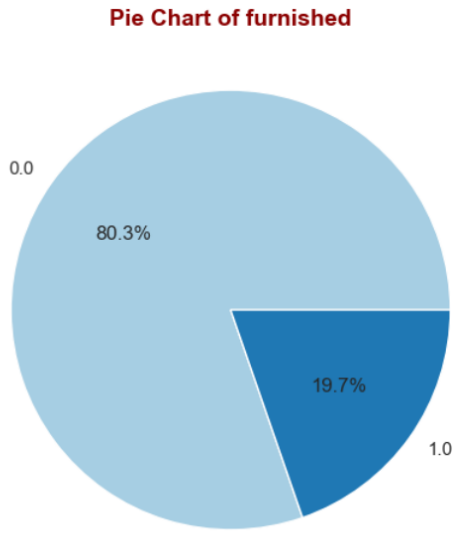


Figure 14

Condition: Univariate Analysis

This is a categorical variable that rates house condition on a scale from 1 to 5. Most houses have a medium rating of 3, followed by a rating of 4, which indicates good condition, and then a smaller number with a top rating of 5, meaning excellent condition. Only a very small percentage of houses are rated below average as shown the bar plot.

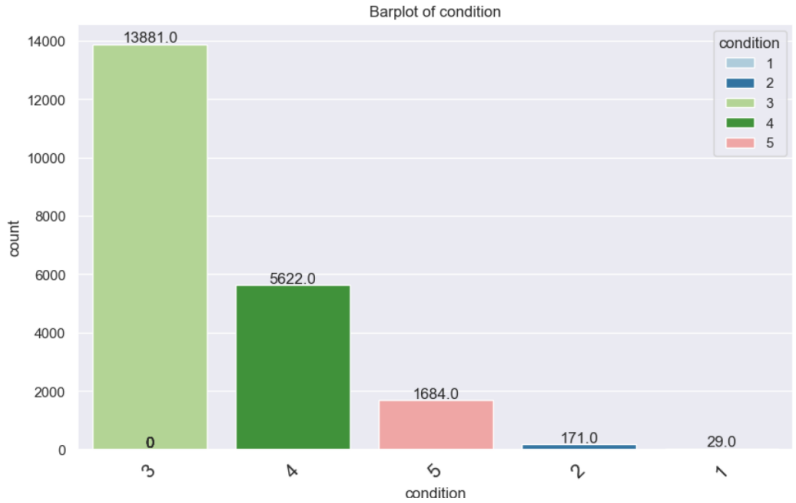


Figure 15

Coast: Univariate Analysis

This shows whether the house has a water front or not, but less than 1% of houses got a water front. Some houses have unknown data related to this.

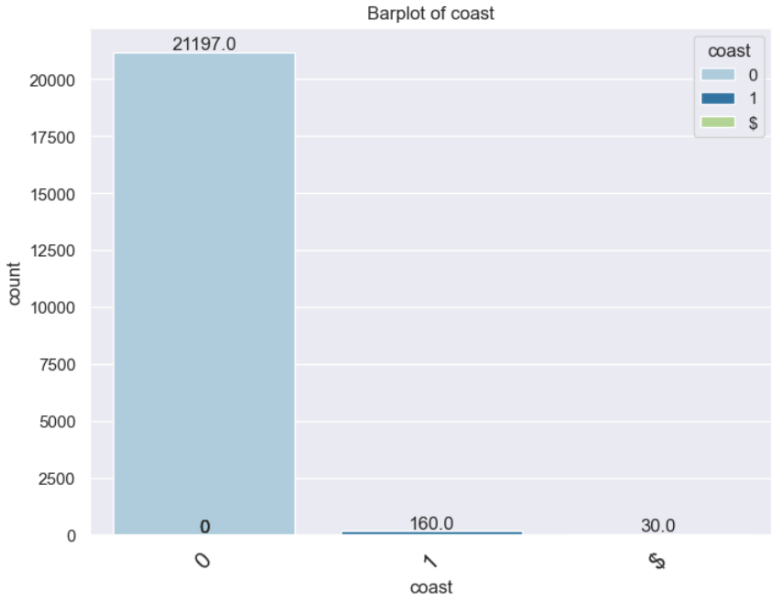


Figure 16

DayHours: Univariate Analysis

67% sales occurred in 2014 and rest in 2015 as shown in following figure.

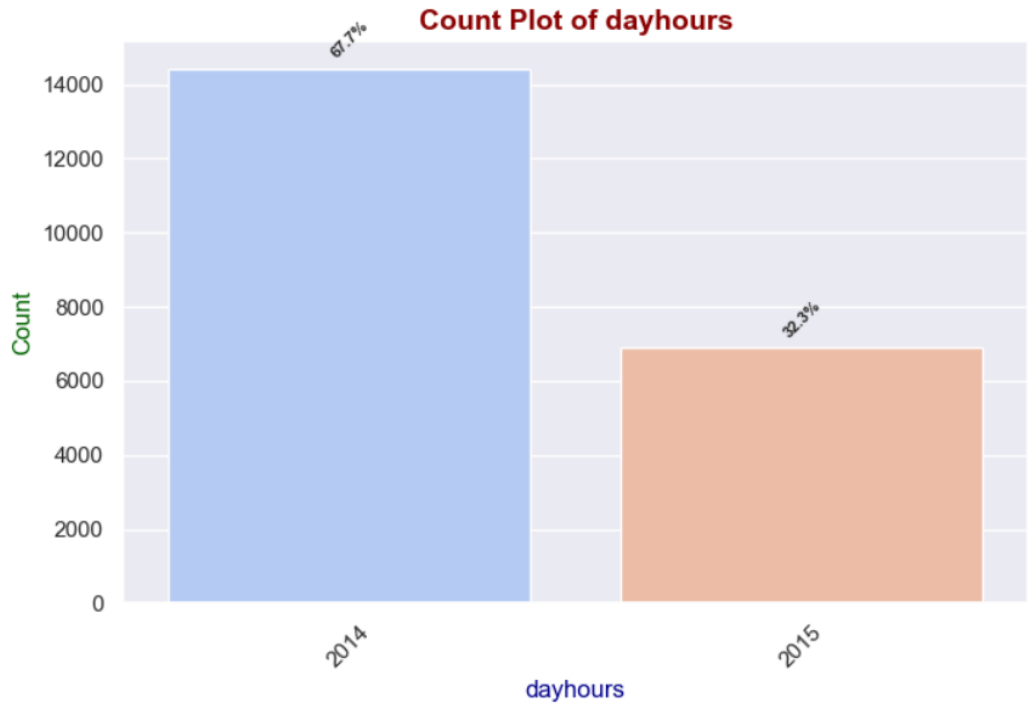


Figure 17:Year wise countplot for number of houses sold

- Highest sales are in the month of May,April, June and July – During months of Spring and Early summer.
- Winter months see a decline in the sales

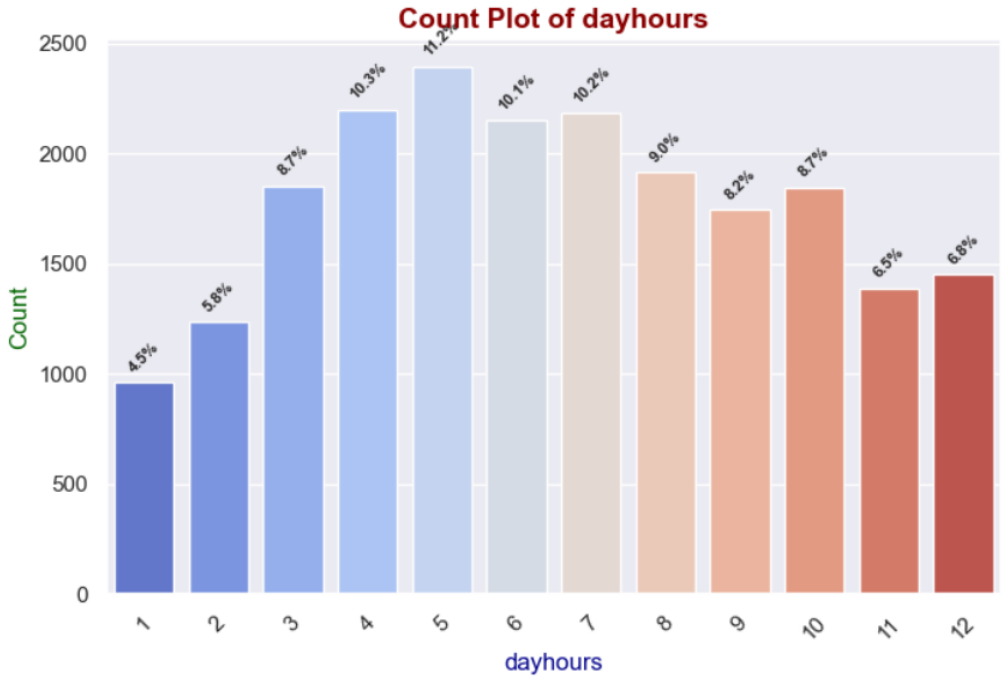


Figure 18:Month wise count of number of houses sold

Latitude and Longitude : Univariate Analysis

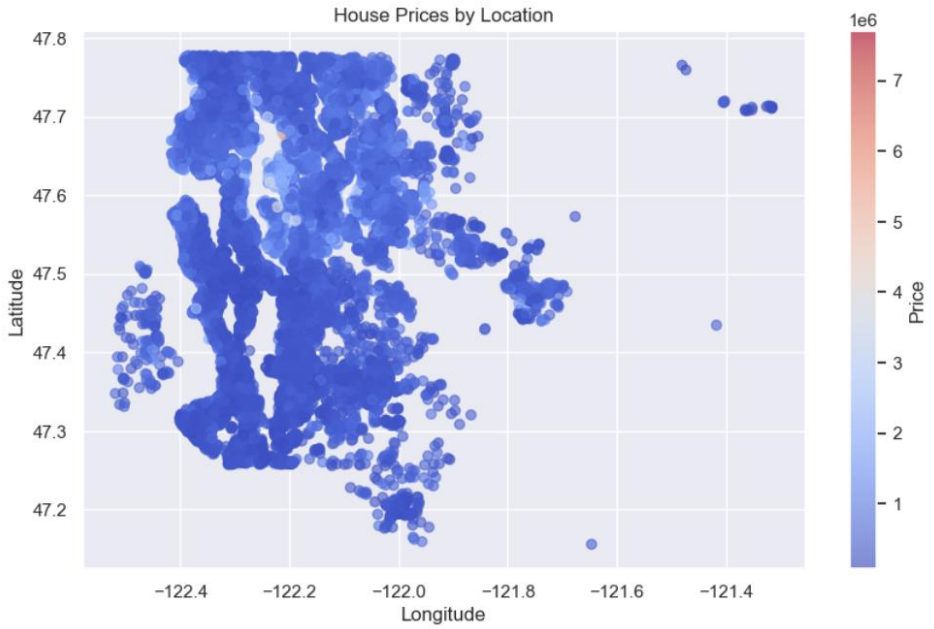


Figure 19:Price per location

The price based location analysis shows that the high priced houses and low priced houses are not concentrating in one particular location but spread around.

Bivariate Analysis

Correlation Matrix

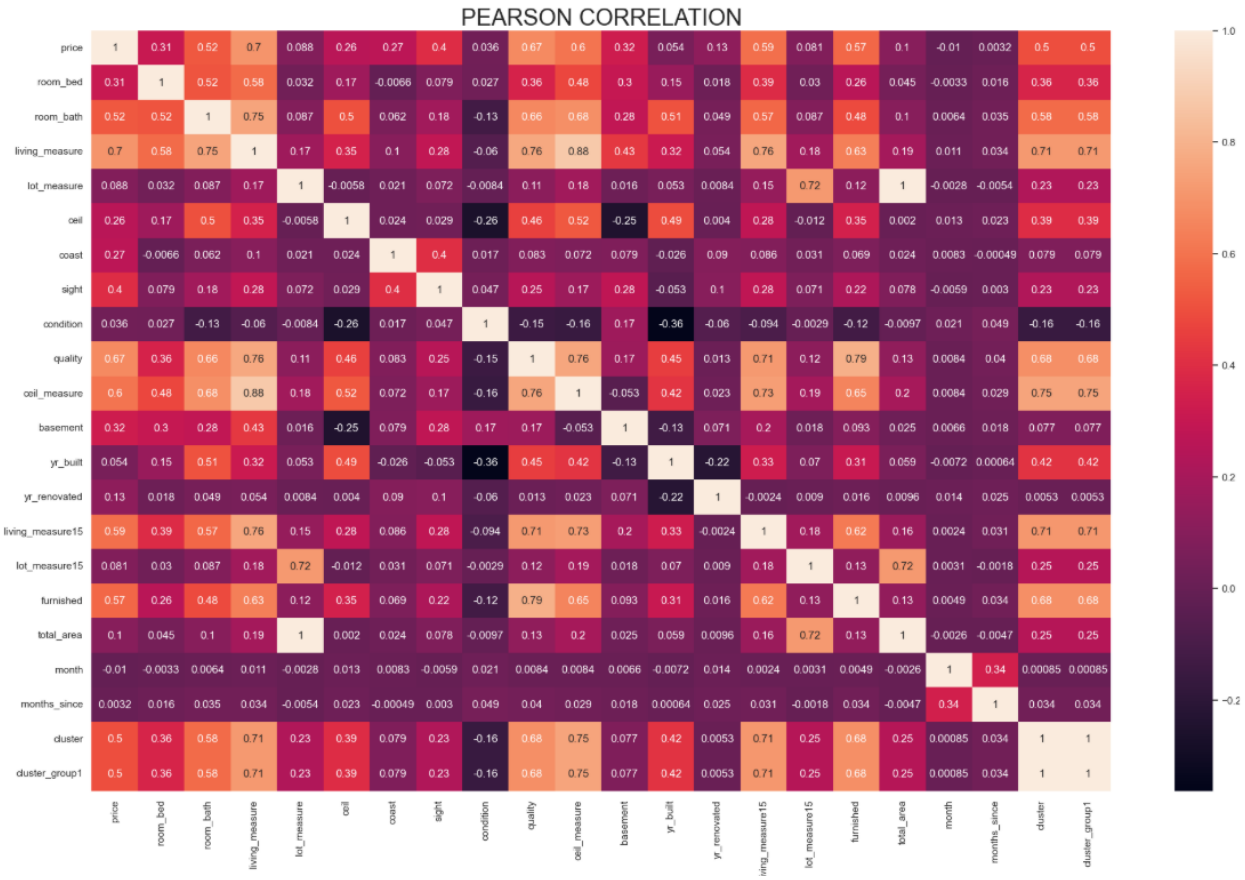


Figure 20:Correlation matrix

House Price — Strongest Influencing Factors

- Price has strong positive correlation with:
- living_measure (0.70)
- quality (0.67)
- ceil_measure (0.60)
- living_measure15 (0.59)
- furnished (0.57)
- room_bath (0.52)

These are the key features that help explain or influence the house price.

Bedroom & Bathroom

- room_bed is somewhat related to price (0.31), but not as strong.
- room_bath is more important — it correlates well with:
    - price (0.52)
    - living_measure (0.75)
    - ceil_measure (0.68)

Size Measures (Living, Ceiling, Total Area)

- living_measure and living_measure15 are very highly correlated (0.76), as expected (original vs post-renovation).
- ceil_measure is strongly related to both living_measure (0.88) and quality (0.46).

- total_area and lot_measure are almost identical (correlation = 0.999), meaning they carry the same information before renovation.
- After renovation, lot_measure15 still relates strongly with total_area (0.716), though slightly less.

Renovation and Construction

- yr_renovated has low correlation with most variables (none above 0.13).
- yr_built has moderate correlation with room_bath (0.51) and ceil (0.49).

Furnished

- furnished is positively correlated with:
  - price (0.57)
  - quality (0.79) — very strong
  - living_measure (0.63)
  - ceil_measure (0.65)

Location: Latitude, Longitude, Zipcode

- lat, long, and zipcode have **very weak correlations** with price and other features.
- zipcode and long have a negative correlation (–0.56), possibly reflecting regional differences.
- Location variables don't appear to influence price much in this dataset.

Other Variables

- coast (waterfront view) has low correlation with price (0.27).
- sight (number of sightings) has moderate correlation with price (0.40).
- basement has weak to moderate correlation with:
  - price (0.32)
  - room_bed and living_measure (~0.30–0.43)

Low or No Correlation

These variables have **minimal influence** based on this correlation matrix:

- condition (very weak correlation with everything)
- yr_renovated (low across all)
- zipcode, lat, long (negligible)
- cid (just an ID, doesn't affect anything)

Price trend

The price trend does not show a clear upward or downward movement over time. Instead there are pretty huge fluctuations indicating variability in house prices. Similarly, the number of units sold remains stable, with a slight dip observed during the early months of 2015.

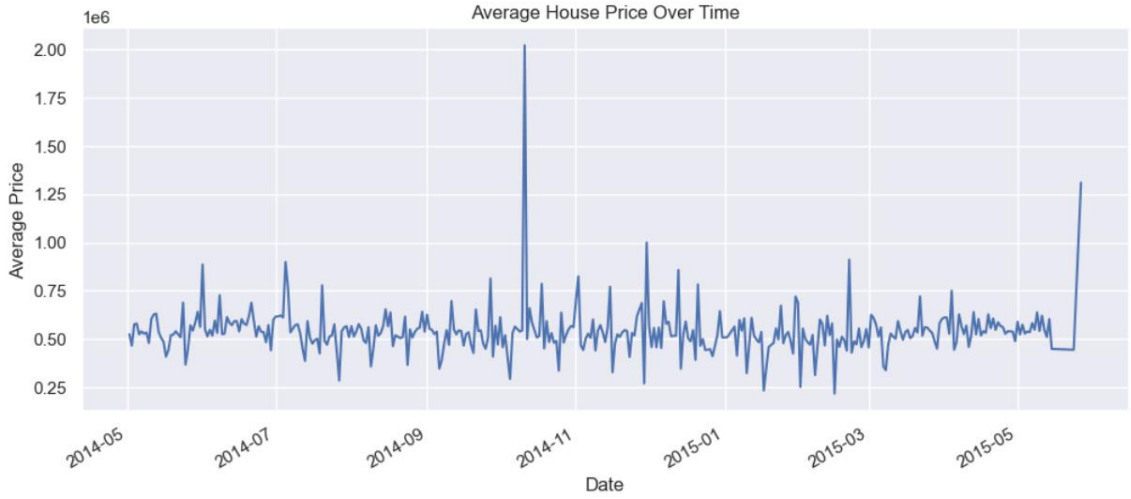


Figure 21

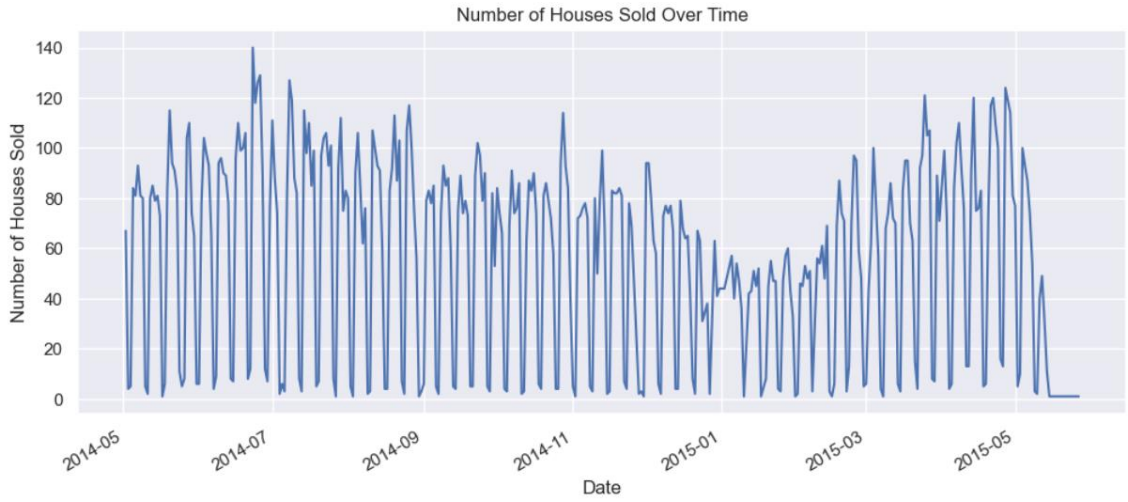


Figure 22

House price vs month

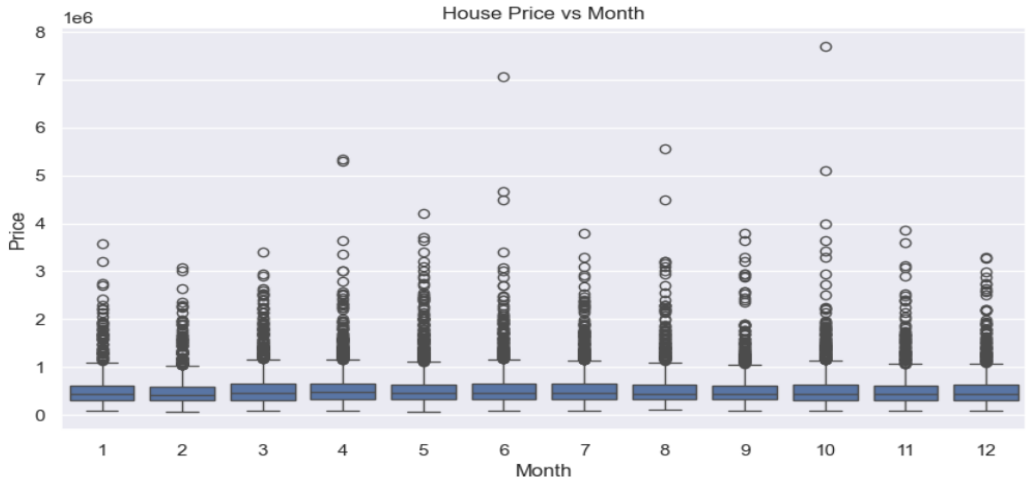


Figure 23

Each month's sales have many outliers in a similar price range. Expensive homes are sold randomly without following any clear pattern. There's no obvious month with noticeably high volume of sales.

House price vs measure type and room count

- It's no surprise that the size of a house plays a big role in its price. The original living space shows the strongest link to how much a house costs.
- Even the living space after renovation shows a similar pattern. After renovations, it's also clear that most homes don't go beyond 6000 sq ft, meaning some properties were reduced in size.
- Some smaller homes gained a lot in value after renovation — this can be seen as scattered high-price points in the plot.
- When we look at lot size, the connection to price isn't as strong. Bigger lots don't always mean higher prices — likely because factors like where the house is, how the land can be used, or legal limits affect the price.

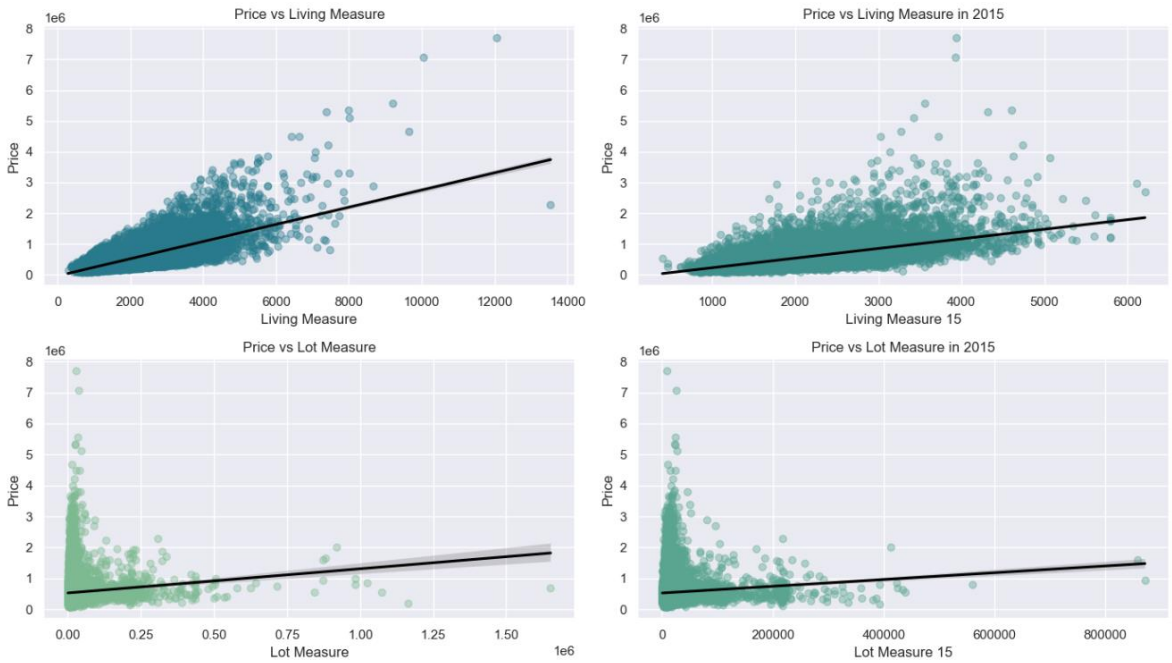


Figure 24:Scatter plot of lot_measure and living_measure

- Having more bedrooms usually means a higher price, but that's not always true. In homes with 3 to 5 bedrooms, there are some that are priced much higher than others. This shows that price depends more on the quality and size of rooms, not just the number of them.
- In case of number of bathrooms, the price is definitely increasing as the number of bathrooms also increase.



Figure 25

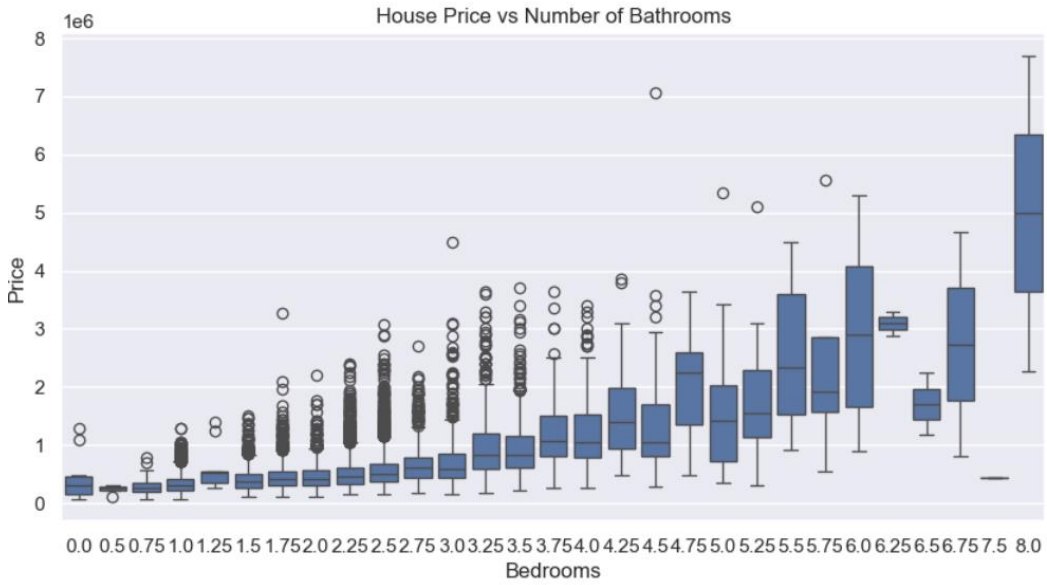


Figure 26

Location based cluster

8 different clusters/regions are identified based on the given locations of sold properties. Following map is showing the location groups

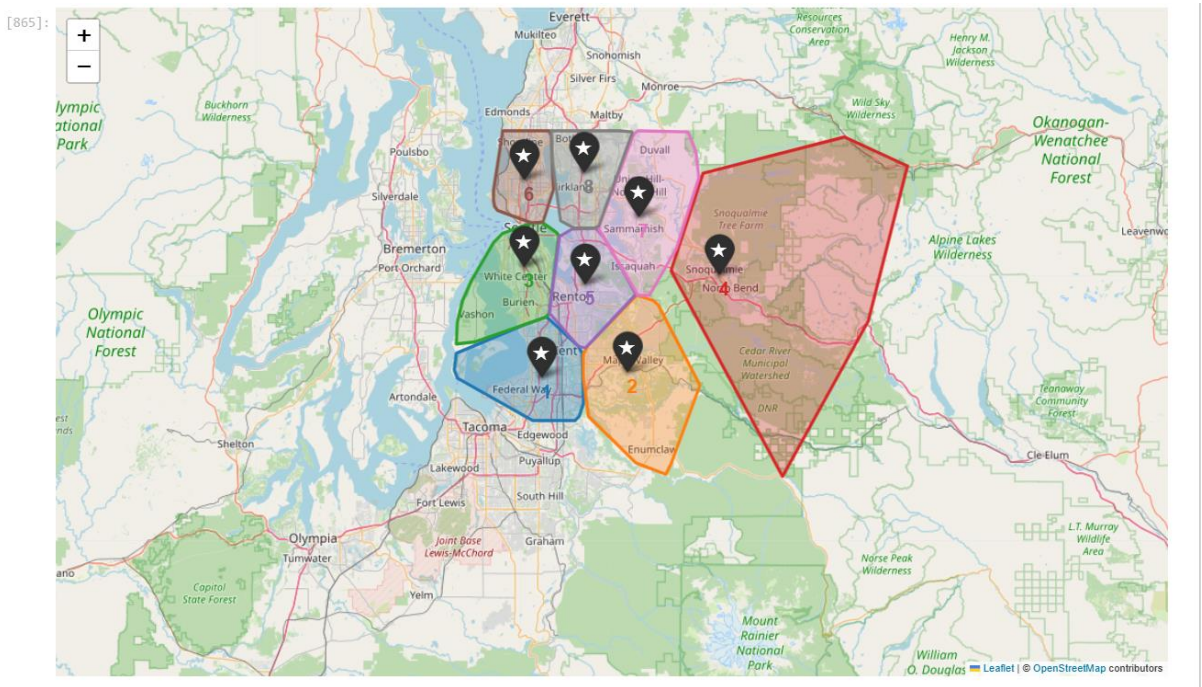


Figure 27: Location cluster

Data Preparation for Clustering

Cluster analysis is a way to group similar houses together based on their features — like price, size, and quality. The goal is to find patterns in the data that can help us understand different types of homes and how they're priced.

To get meaningful results, a few important steps were followed before clustering:

1. **Removing Unnecessary Columns**: We dropped columns like ID, zipcode, dayhours and other details that are not very relevant in understanding the house characteristics.

2. **Feature Engineering and Encoding Categorical Data/Ordinal data:** Since machine learning works best with numbers, we have to convert certain categorical and raw features into numerical ones if possible. First we have converted text-based features (like whether a house is on the coast or if furnished) into a format the model can understand which is known as Encoding Categorical Data.

```
Index(['price', 'room_bed', 'room_bath', 'living_measure', 'lot_measure',
       'ceil', 'sight', 'condition', 'quality', 'ceil_measure', 'basement',
       'yr_renovated', 'living_measure15', 'lot_measure15', 'total_area',
       'months_since', 'location_cluster', 'age_yr_built', 'coast_1.0',
       'furnished_1.0'],
      dtype='object')
```

Table 12:Result of converted categorical variables

Following are the newly engineered features which is a numerical representation of certain raw features features:

- Dayhours was an abstract date object that just mentioned the date of sale. It is mapped to a numerical variable 'month_since' which is numerical variable.
- 'year_built' is converted to a numerical variable called 'age_ye_built' that is representing age of the house.
- 'Location_cluster' is a new variable created from 'lat' and 'long' that is a cluster of 8 different locations

3. **Scaling Numbers**: Some features like square footage have large values, while others like the number of bathrooms are much smaller. We scaled them so all features are treated fairly in the analysis. Following table is an example after scaling:

	price	room_bed	room_bath	living_measure	lot_measure	ceil	sight	ceil_measure
0	600000	0.677059	-0.473899	1.057393	-0.136743	-0.915970	-0.305766	0.013561
1	190000	-1.473693	-1.449847	-1.536902	-0.289876	-0.915970	-0.305766	-1.352287
2	735000	0.677059	0.827364	1.046493	-0.306448	0.934629	4.916381	1.512368
3	257000	-0.398317	0.502048	-0.370559	-0.274899	0.934629	-0.305766	-0.058961
4	450000	-1.473693	-1.449847	-1.046384	-0.253906	-0.915970	-0.305766	-0.808365

Table 13:Glimpse of scaled variables

4. **Choosing the Most Important Features:** We used a method called Recursive Feature Elimination (RFE) to select only the top features that influence the model. This helps keep the clustering focused and efficient. This method selected 15 such features based on this method: ['room_bath', 'living_measure', 'lot_measure', 'sight', 'quality', 'ceil_measure', 'basement', 'living_measure15', 'lot_measure15', 'total_area', 'months_since', 'location_cluster', 'age_yr_built', 'coast_1.0', 'furnished_1.0']

5. **Clustering Method:** Because we had both numbers and categories which are converted to numerical type, we used a method called K-Prototypes, which handles mixed data types well.

6. **Finding the Right Number of Groups:** To decide how many clusters made the most sense, we used techniques like the Elbow Method and Silhouette Score.

```
k=2, silhouette=0.292
k=3, silhouette=0.274
k=4, silhouette=0.240
k=5, silhouette=0.171
k=6, silhouette=0.172
k=7, silhouette=0.181
k=8, silhouette=0.155
k=9, silhouette=0.169
```

Table 14:Silhouette score for different k values or number or clusters

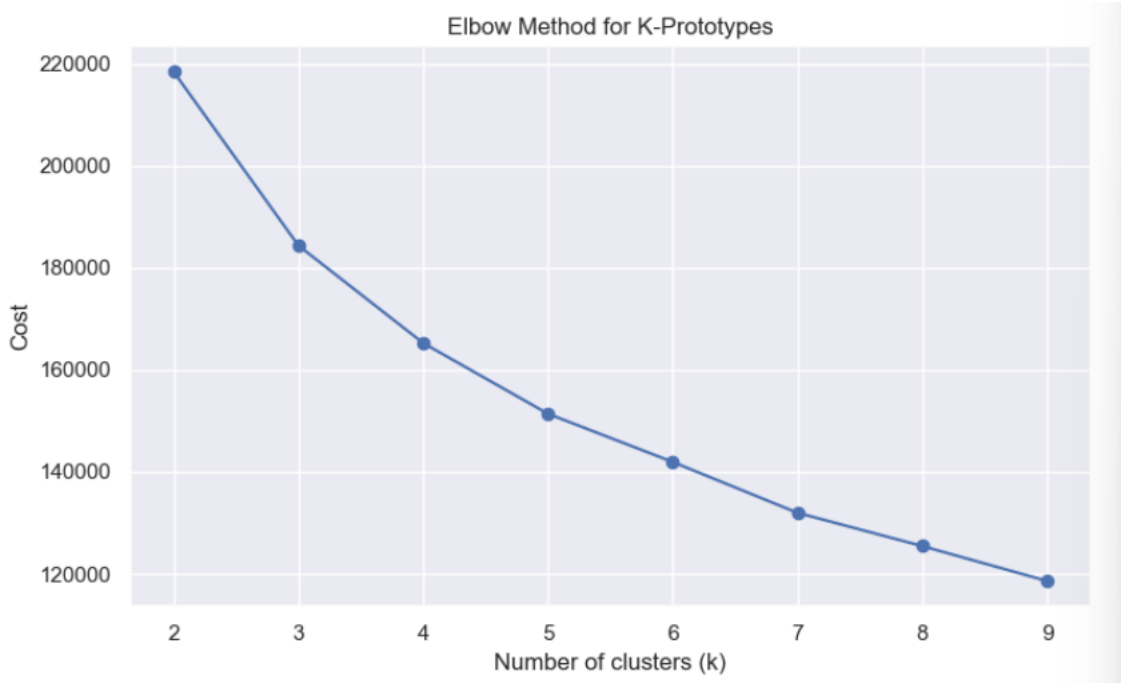


Figure 28:Elbow plot to check the number of clusters, k=2 or or 4 can be good values

Based on the two methods, the number of clusters to go for is 3 and thus three clusters are created. Further details are in following sections.

Cluster Profiling:

Cluster 0 → High-price, large houses in small to medium lots, executive to premium quality rating, exclusive coastal homes, multi-level houses and many houses comes furnished

Cluster 1 → Medium-price, medium sized houses in large lots, standard quality rating, smallest cluster

Cluster 2 → Low to medium-price, low to medium sized houses in medium sized lots, standard quality ratings and mostly unfurnished, budget friendly options

---

Cluster 0 Summary

- **Price:** Avg ~ $852775, wider spread (std ~ $509k), max price very high ($7,700,000)-
- **Rooms:** Average rooms around 3-4 bedrooms and ~3 bathrooms
- **Living area:** Larger (mean living_measure ~ 3140 sq ft, after renovation ~ 2756 sq ft)
- **Lot size:** medium lot sizes – smaller than cluster 1, on average the lot size is ~15k
- **Quality:** Most houses have higher quality ratings between 9 to 10
    - Quality 9.0 (38%), 8.0 (31%), 10.0 (18%), 11.0 (6%)
    - Minimum quality rating is 6 to goes up to 13
- **Condition:** Mostly condition 3 (about 75%), then 4 (18%), very few in worse conditions
- **Furnishing:** 64% houses are furnished
- **Other:** Mostly medium to high conditions and higher quality, likely more expensive, bigger houses, coastal view houses are only in this cluster

---

Cluster 1 Summary

- **Price:** Avg ~ $666,437, medium spread ~351k, max price ~ $2,280,000
- **Rooms:** Average bedroom count similar to cluster 0 i.e., 3-4, average bathroom is ~2
- **Living area:** small to medium size homes mostly with average living measure of 2878 sq ft.
- **Lot size:** Large lot sizes with average size of 254k sq ft.
- **Quality:** Quality mainly between 6-8
    - Quality 7.0 (26%), 9.0 (22%), 8.0 (21%) dominate
    - Minimum quality rating of 4 or 5, range of ratings from 4 to 12
- **Condition:** Condition mostly 3 (70%), but more in better, condition 4 – 21%, condition 5 – 4%
- **Furnishing:** 56% are unfurnished
- **Ceil:** 2 floors: 75%, 1 floor: 20%
- **Other:** Smaller, mid-quality homes, lower prices, generally good conditions but not as high quality as cluster 0

Cluster 2 Summary

- **Price:** Avg ~ $413,483, lower spread (std ~ $170,730), max price ~ ₹1,550,000

- **Rooms:** Generally smaller homes (fewer bedrooms/bathrooms compared among all the clusters) On average 3 rooms and 1-2 bathrooms
- **Living area:** Smaller to medium sized homes (mean living_measure ~ 1616 sq ft
- **Lot size:** Smaller lots overall – average lot size of 8k sq ft.
- **Quality:** Quality mainly between 6 to 8
    - Quality 7.0 (56%), 8.0 (27%), 6.0 (13%) dominate
    - Fewer homes with quality 9 or above, range of rating from 1 to 10
- **Condition:** Condition mostly 3 (60%), condition 4 – 29%, condition 5 -8%
- **Furnishing:** 98% unfurnished
- **Ceil:** 1 floors: 61%, 2 floor: 24%
- **Other:** Smaller, mid-quality homes, lower prices, generally good conditions but not as high quality as cluster 0

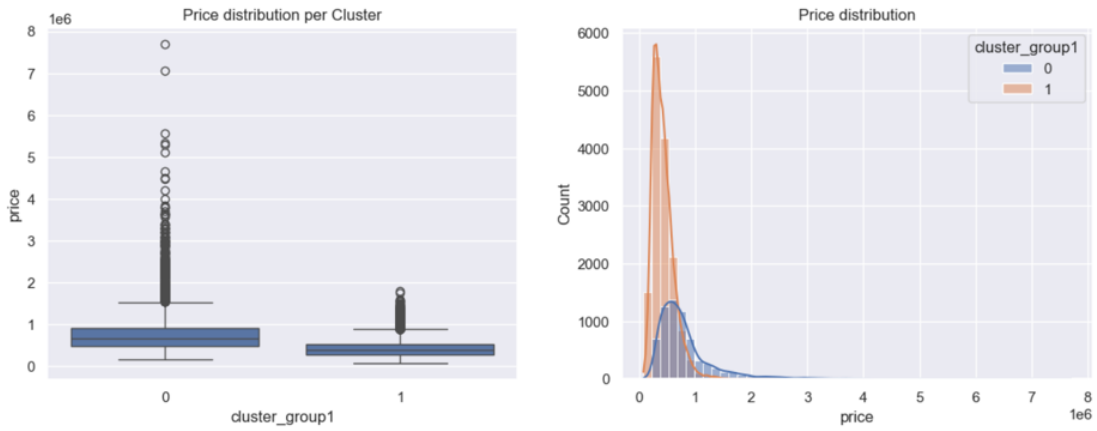


Figure 29:Price comparison of clusters-The price variation of cluster zero is highest compared to cluster 1

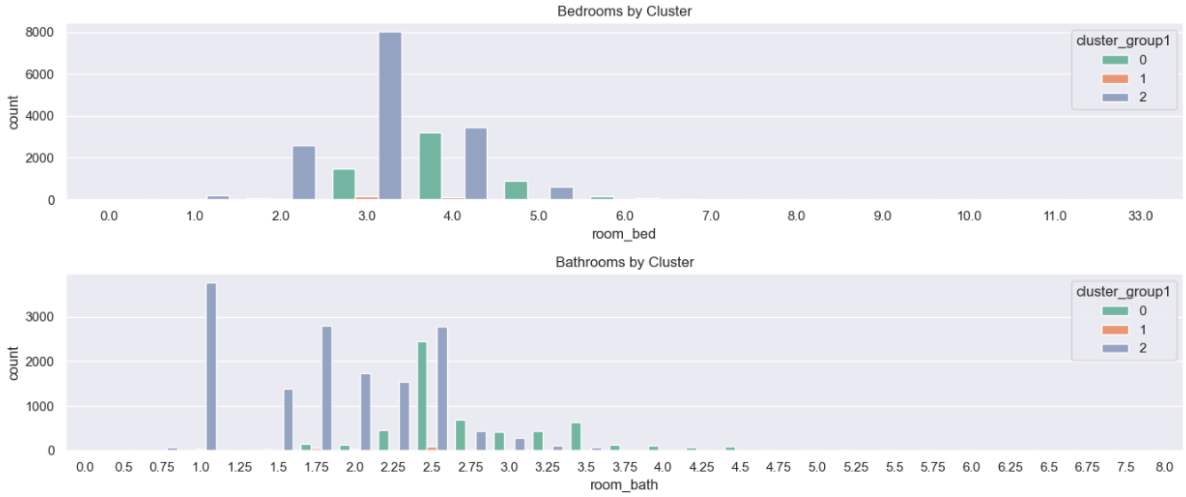


Figure 30:This figure is showing the comparison of bedroom and bathrooms per house. Cluster 0 has most houses with room count of 4 or 3 or 5. For cluster 1, the count or rooms is mostly with 3 or 4, Cluster 1 has rooms around 3. So this is indicating cluster 0 has houses with more rooms, followed by cluster 1 and cluster 2

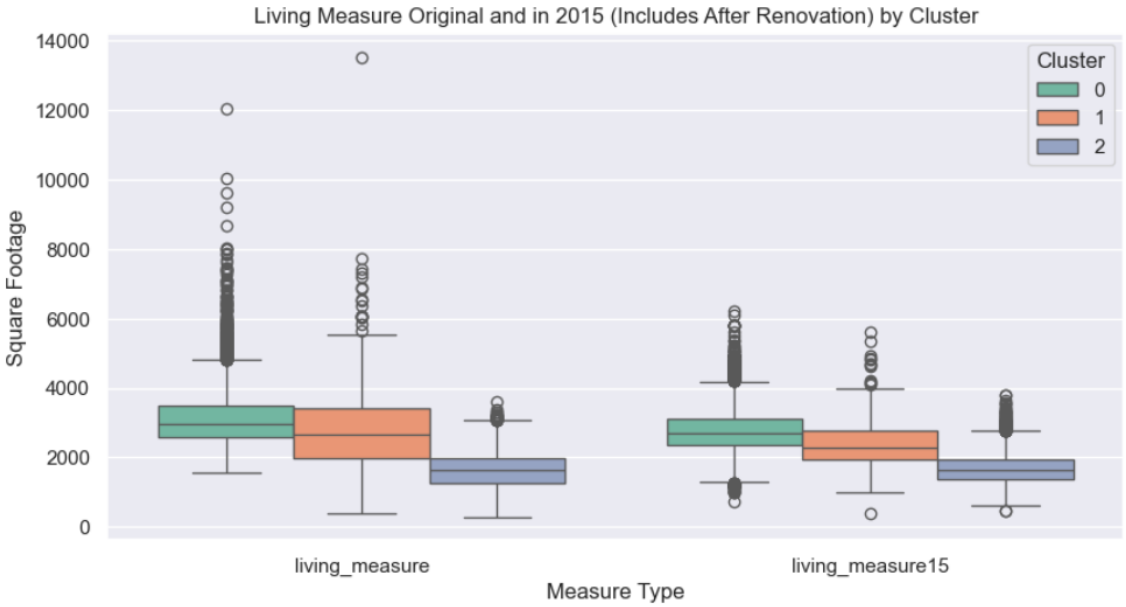


Figure 31:This is comparing the living_measure original and as of 2015 for each cluster. The living measure of cluster 0 is the highest followed by cluster 1. The smallest size is for cluster 2. The size of cluster 0 has many outliers in the original one. After 2015 , many outliers has reduced for cluster 0. Similalry the cluster 1 and cluster 2 also has a shrinkage in the living measure in 2015 compared to original measure indicating that the renovations resulted in reducing original house dimension.. This shows there are renovated building in all the clusters

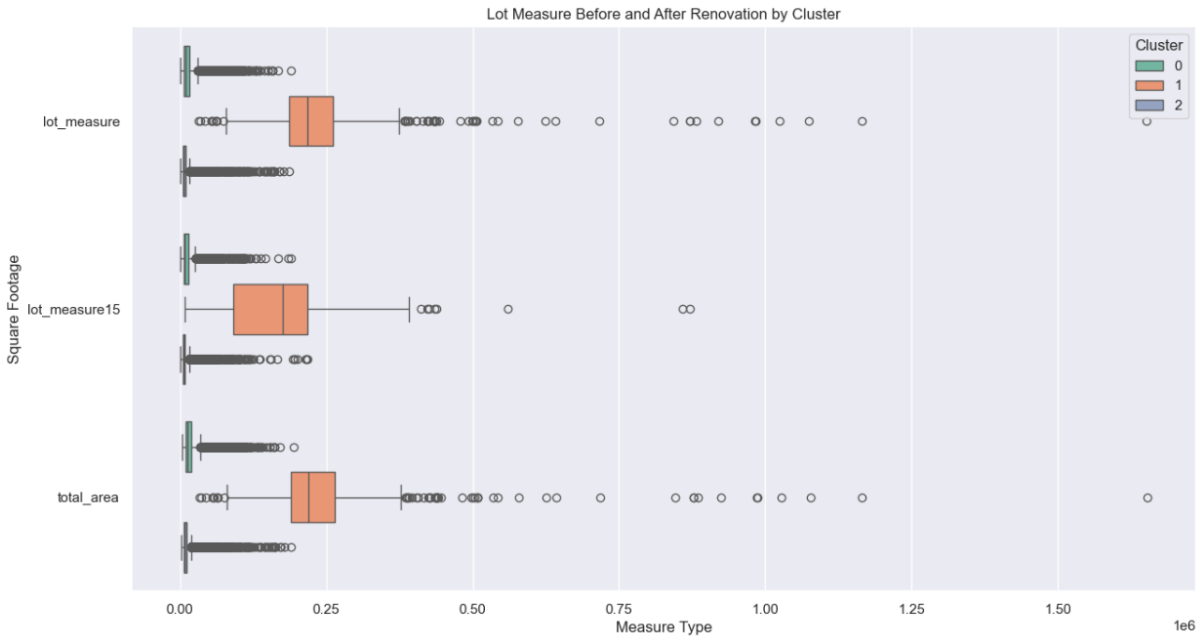


Figure 32:This figure is comparing lot_measure with lot_measure15 and total_area for both clusters. Cluster 1 has highest lot measure size and variations. The lot size of cluster 0 and cluster 2 is smaller then cluster 1 and is of similar range.

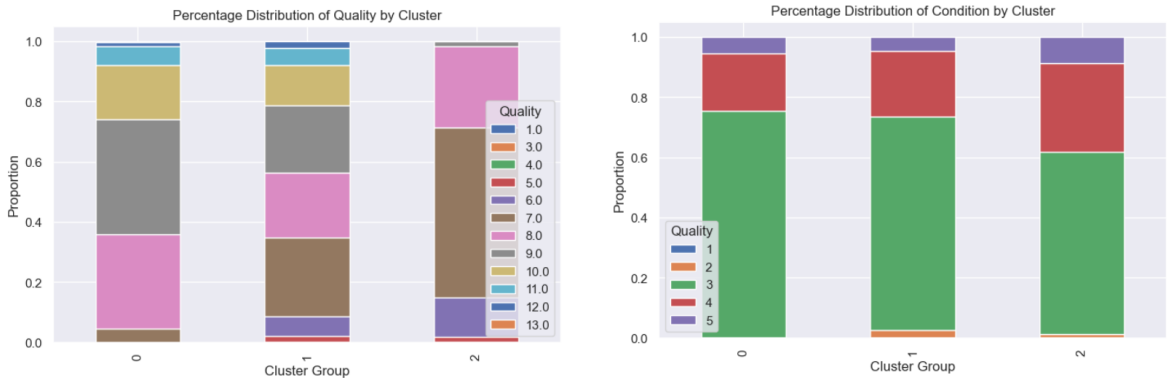


Figure 33:This is showing percentage distribution of the quality and condition of houses. Cluster 0 has more percentage of high quality rated houses-ratings of 8, followed by 9,10 and 7. While in case of cluster 1, medium quality rated houses can be seen – majority has a rating of 7, followed by 8,6. Finally in case of cluster 2, the houses have mainly standard rating of 7. So among the clusters, cluster 0 has higher quality houses comparatively. In terms of condition all the clusters have majority houses with medium rating of 3 followed by rating of 4 and 5. The proportions of the condition rating vary slightly between clusters but this is not a distinguishing feature for the clusters

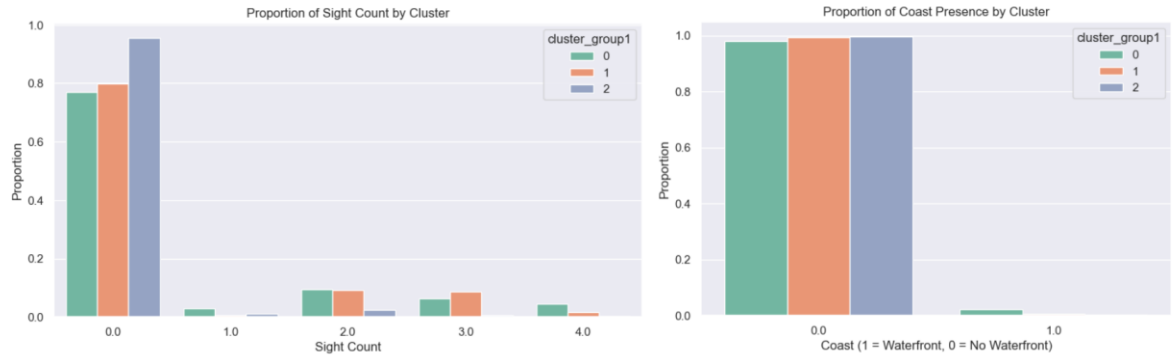


Figure 34:This is comparison of coasts and sights. Majority houses have no waterfront or sighting- But it can be noted that houses with coast is only present in cluster 0.

Insights from EDA and recommendations

- The data is unbalanced in several places-a large portion of houses are not renovated, most homes are in average or good condition and very few in poor or excellent condition, majority houses in certain zip codes or quality ratings which can dominate the analysis. Further the data is for a limited time period of year. As imbalance can bias models or analyses toward majority class, we can try sampling techniques such as **oversampling** the minority class or **undersample** the majority. It is also better to have a large dataset that has multiple years of data to get more insights and also for prediction analysis
- Cluster 0 has large and expensive houses with higher quality ratings. These could be marketed as premium properties. Houses with coastal view can be given attractive headlines for marketing and also a price analysis might be needed to match the price with real market demand
- Cluster 1 contains medium sized and more affordable houses but in bigger lot sizes. Renovation of such properties can also be considered.
- Cluster 2 has small to mid-sized homes in smaller lots. These homes may appeal budget-conscious customers. These are the most sold properties. Here also a pricing analysis needed to reduce undervaluing houses. The smarter layouts and multi-functional spaces can be highlighted while marketing.

Model Building

In order to predict house price, let's go ahead with building models that can predict prices. As a part of this the plan is to build multiple models, compare the performances and choose the one that best performs for house price prediction. Following are 8 such models:

1. Linear Regression
2. Lasso Regressor
3. Ridge Regressor
4. Polynomial Regressor
5. Random Forest Regressor
6. SVM Regressor
7. XGBoost Regressor
8. AdaBoost Regressor

Among the 8 model, the first four are parametric models that is , such models make strong assumptions about the shape of the data (like "it follows a line or a curve") and then try to fit the best version of that shape. So while building the models these assumptions also need to be checked.

Metrics for evaluation:

To evaluate and compare the model performances following 5 metrics are being used:

Metric Name	Acronym	Simple Definition
R-squared	R^2	Tells how much of the variation in the target variable is explained by the model.
Adjusted R-squared	Adj. R^2	Similar to R^2 but adjusts for the number of features used; discourages overfitting.
Root Mean Squared Error	RMSE	Measures how far predictions are from actual values, giving more weight to large errors.
Mean Absolute Error	MAE	Average of all prediction errors, treating all errors equally.
Mean Absolute Percentage Error	MAPE	Shows average prediction error as a percentage of the actual value.

Other evaluation methods:

Apart from numeric metrics, we also **visualize residuals** which are the differences between actual and predicted values to check how well the model is performing.

- If the residuals are randomly scattered around zero, it means the model is doing well.
- If we see patterns or curves, the model might be missing something like non-linearity or variable interactions.
- If residuals increase or decrease as predictions grow, it may indicate heteroscedasticity (changing variance), which violates linear regression assumptions.

Data Preparation for model building:

1.**Fixing Missing Values**: Sometimes data is incomplete. We either fill in the blanks with a value or remove those rows so the model doesn't get confused.

2.**Converting all features to numerical data type**: Machine learning models only understand numbers. So, we convert categories like "Yes"/"No" or "Type A"/"Type B" into numbers using special techniques.

3.**Scaling**: Some columns have large numbers, others small. We adjust the scale so they all play fair. This helps certain models work better.

4.**Splitting the Data**: We split the dataset into two parts: one to train the model and another to test how well it works.

5.**Outlier check:** We sometimes fix or remove them if they affect results too much. But in this case outliers are left untouched

6.**Choosing the Right Features** : Not every column helps the model. We pick the most useful ones to keep things simple and focused. This is optional step. But for model building this step is skipped.

Model 1: Linear Regression

Linear Regression is a parametric model. It assumes the relationship is a straight line:

$$Price = a * Size + b$$

The model just needs to learn the values of a and b.

Before proceeding with model building following assumptions are to be checked :

Assumptions of Linear Regression:

Linear regression works best when certain conditions are met:

1. **Linearity**: The relationship between input features and the target should be roughly a straight line. To check this Plot the *residuals* (errors) against the predicted values or against each predictor. If points scatter randomly around zero, then the linearity is good. If it is a curve or trend then linearity is violated.
2. **No Multicollinearity**: The input features should not be highly correlated with each other. If the predictors have collinearity, then the coefficients given by the model is not reliable. Multicollinearity can be checked by variance inflation factors (VIF) of each predictor and those that have some kind of collinearity appears with high VIF factor. Such variables need to be removed for linear regressor to work properly. High VIF variables are iteratively removed based on its effect on the model building.

	features	VIF			
0	const	1.703350	13	lat	1.122686
1	room_bed	1.635891	14	long	1.499251
2	room_bath	3.361109	15	living_measure15	2.890290
3	living_measure	inf	16	lot_measure15	2.055095
4	lot_measure	inf	17	total_area	inf
5	ceil	2.018995	18	months_since	1.008251
6	sight	1.418982	19	coast_1.0	1.193539
7	condition	1.237491	20	furnished_1.0	2.794925
8	quality	4.801571			
9	ceil_measure	inf			
10	basement	inf			
11	yr_built	2.399758			
12	yr_renovated	1.143424			

Table 15: VIF factor of various features in the dataset. Those with VIF factor more than 5 are iteratively removed from the model building after checking its effect on RMSE and adj R2. In this case the living_measure, lot_measure, ceil_measure, basement, total_area is having VIF of infinity(inf) and they were iteratively removed. Sometimes after removing a particular variable, the VIF of other variables can change to a lower value and need not to removed

3. **Homoscedasticity**: The spread of errors (residuals) should remain consistent across predictions or in other words constant variance. This can also be checked by plotting the residual and by doing a Goldfeld-Quandt test.

In the initial test, the golfdfeld-Quandt test failed, showing that the dataset had heteroscedasticity. This can be removed by some kind of data preprocessing such as log transformation of target or predictor variable, exponential transformation. **Here log transformation of target variable helped to remove heteroscedasticity**. This was confirmed by the goldfield-quandt test.

The goldfeldquandt test is a statistical test that checks the probability for a null hypothesis. If we get a p-value > 0.05 we can say that the residuals are homoscedastic. Otherwise, they are heteroscedastic.

- Null hypothesis: Residuals are homoscedastic
- Alternate hypothesis: Residuals have heteroscedasticity

The final test result is ('F statistic', 1.00), ('p-value', 0.40)]. **Since p-value > 0.05, we can say that the residuals are homoscedastic. So, this assumption is satisfied.**

4. **Normality of Residuals**: The differences between actual and predicted values should be normally distributed. This can be checked visually by plotting the histogram of residuals and on top of that Shapiro-wilk test

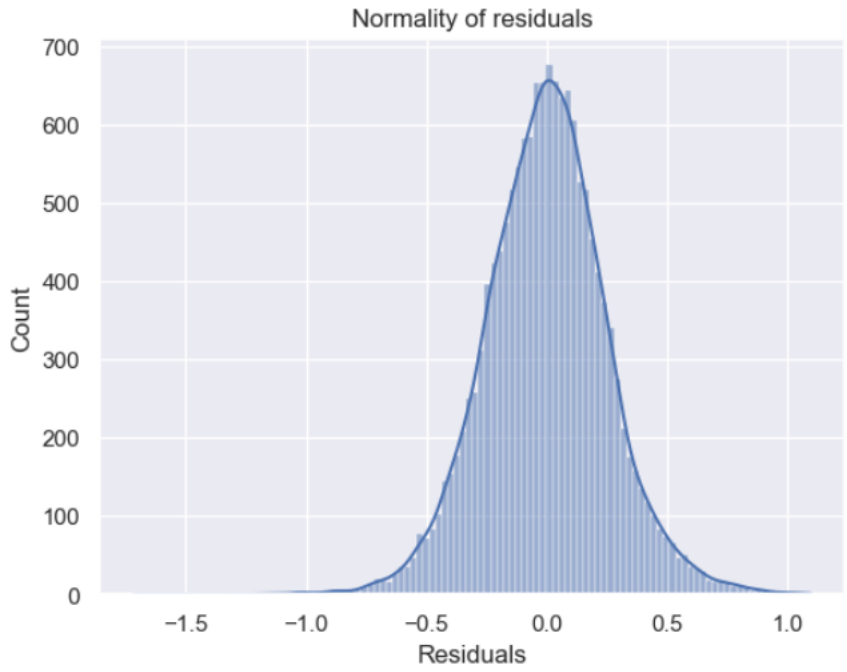


Figure 35: Histogram of residuals for linear regressor model

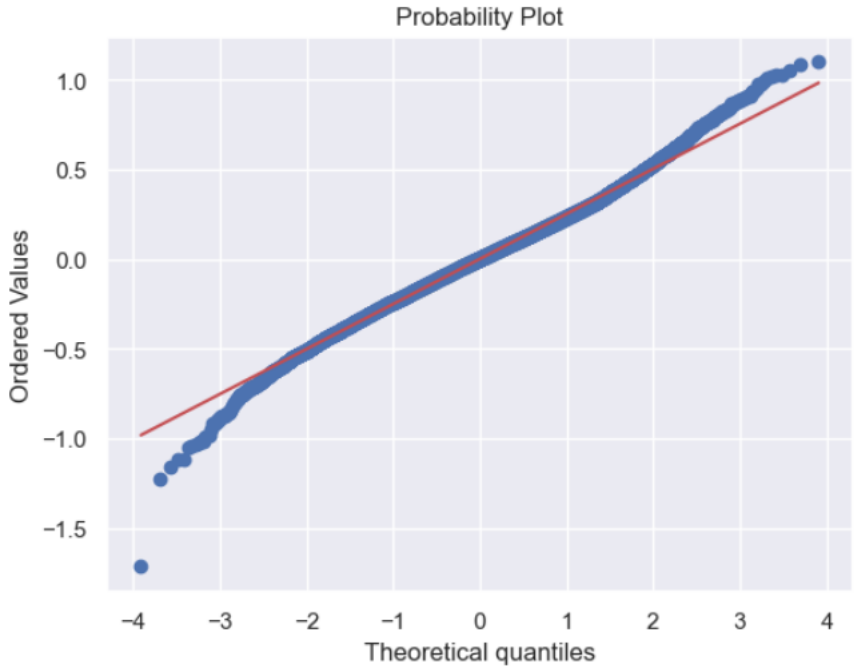


Figure 36:Q-Q plot of residuals for Linear regression model

Shapiro wilk also considers to hypothesis and finally check the probability or p-value of the null hypothesis.

- Null hypothesis: The data is normally distributed
- Alternate hypothesis: The data is not normally distributed.

Shapiro Result for the model: (statistic=0.99, pvalue=1.38e-23).

Since p-value < 0.05, the residuals are not normal as per the Shapiro-Wilk test. Strictly speaking, the residuals are not normal.
However, as an approximation, we can accept this distribution as close to being normal as observed in the distribution plot of residuals. Statistic is 0.994, which is very close to 1, suggesting almost normal shape. So, the assumption is satisfied.

5. **Independence of Errors**: The residuals should not be correlated with each other. This is checked via Durbin-Watson test (numbers close to 2 mean “independent”).

Residual Analysis:

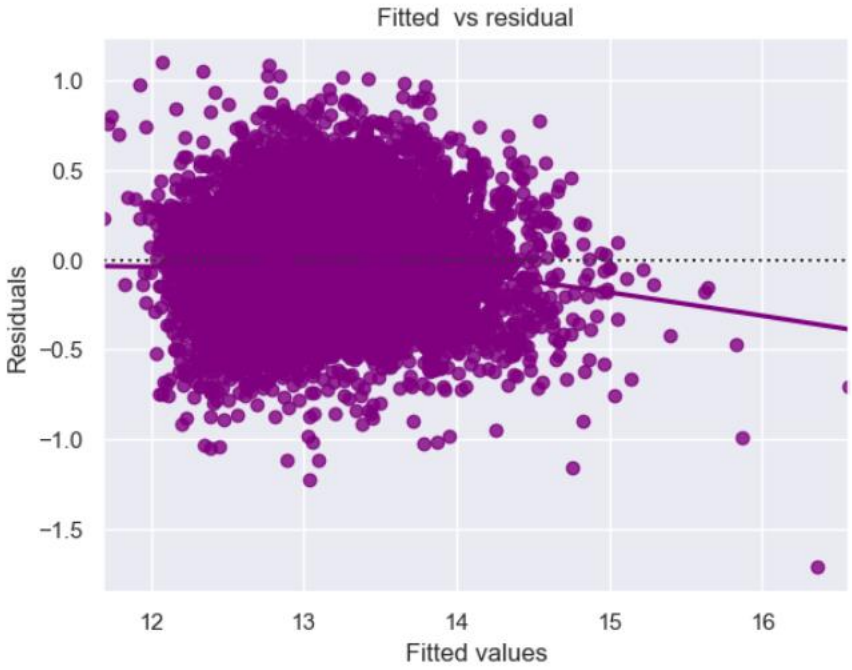


Figure 37: Residual plot for linear regression model

The residuals seem to be uniformly distributed around zero in this case. There are particular patterns observed here.

OLS Regression Results

Dep. Variable:	price	R-squared:	0.649
Model:	OLS	Adj. R-squared:	0.649
Method:	Least Squares	F-statistic:	2125.
Date:	Wed, 23 Jul 2025	Prob (F-statistic):	0.00
Time:	13:22:40	Log-Likelihood:	-3622.5
No. Observations:	14926	AIC:	7273.
Df Residuals:	14912	BIC:	7380.
Df Model:	13		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	13.0458	0.003	5142.615	0.000	13.041	13.051
room_bed	-0.0232	0.003	-7.294	0.000	-0.029	-0.017
room_bath	0.0644	0.005	14.059	0.000	0.055	0.073
living_measure	0.1261	0.006	22.376	0.000	0.115	0.137
ceil	0.0503	0.003	15.802	0.000	0.044	0.057
sight	0.0356	0.003	12.050	0.000	0.030	0.041
condition	0.0292	0.003	10.419	0.000	0.024	0.035
quality	0.2285	0.005	49.959	0.000	0.220	0.237
yr_renovated	0.0075	0.003	2.782	0.005	0.002	0.013
living_measure15	0.0690	0.004	16.719	0.000	0.061	0.077
months_since	-0.0186	0.003	-7.331	0.000	-0.024	-0.014
location_cluster	-0.0366	0.003	-14.177	0.000	-0.042	-0.032
age_yr_built	0.1583	0.004	43.973	0.000	0.151	0.165
coast_1.0	0.3661	0.033	11.155	0.000	0.302	0.430

Omnibus:	89.489	Durbin-Watson:	2.001
Prob(Omnibus):	0.000	Jarque-Bera (JB):	93.764
Skew:	-0.170	Prob(JB):	4.36e-21
Kurtosis:	3.189	Cond. No.	26.1

Figure 38: Linear regression model summary

Performance metrics:

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.308436	0.246544	0.649422	0.649093	1.897172

Testing performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.309423	0.246251	0.671516	0.670795	1.896439

Table 16

- The model is able to explain ~64 to 67% of the variation in the data
- The train and test RMSE and MAE are low and comparable. So, our model is not suffering from overfitting
- The MAPE on the test set suggests we can predict within 1.8% of the house price
- So the model prediction works almost similar on seen and unseen data and is almost good performance.

Model Interpretation

Model can be interpreted based on the coefficients of each feature takes as shown in the model summary figure 37. There are features with positive and negative coefficients. Following are such **top 5 features that positively affect the price as per the model:**

- **coast_1.0** (0.3661): Houses located near the coast have, on average, a 0.36 unit higher log price compared to inland houses.
- **quality** (0.2285): As the overall quality of a house increases by one unit, the log price increases by about 0.22 units. Higher quality homes are clearly valued more by buyers.
- **Age_yr_built** (0.1583): For every unit increase in age, the house price tends to increase. This indicates that older houses are more expensive.
- **living_measure** (0.1261): An increase in the interior square footage of the house leads to higher prices. Spacious homes increases price.
- **living_measure15** (0.0690): If the living measure after renovations increases, the house price also increases. But this is not as influencing as the original living measure prior to renovations

Top 3 factors that decrease house prices the most:

- **location_cluster** (-0.0366):Location wise clustering has a negative price influence indicating certain clusters has lower prices than others
- **room_bed** (-0.0232): Surprisingly, having more bedrooms slightly reduces the house price, possibly because it reduces other valuable space.
- **months_since** (-0.0186): Houses that have been on the market for longer periods tend to sell for less.

Model 2: LASSO Regression

Least Absolute Shrinkage and Selection Operator (LASSO) regression is a type of linear regression that also helps with feature selection. It works by slightly reducing (or shrinking) the importance of less useful features, and sometimes even setting their impact to zero. This helps build a simpler and more efficient model, especially when there are many features involved. It's especially useful when we want to avoid overfitting and focus only on the most important variables.

Lasso shares the same basic assumptions as linear regression and testing almost similarly:

1. **Linearity** – The relationship between independent variables and the target should be linear.
2. **No or low multicollinearity** – The independent variables should not be highly correlated with each other. Lasso helps handle multicollinearity by selecting one among correlated features, but extreme cases may still be unstable.
3. **Homoscedasticity** – The variance of errors should be constant across all levels of input features. This can be checked by plotting the residuals vs predicted values. Golf Quandt test is not needed here.
4. **Normality of residuals** – The errors (residuals) should be roughly normally distributed.
5. **Independence of errors** – The residuals should not be dependent on each other.

Following are the test and train metrics, residual analysis and observations on meeting the assumptions:

Training performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	192404.876866	121837.704672	0.701925	0.701505	24.477568

Testing performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	219137.410317	128878.805557	0.691117	0.690099	25.295726

Table 17: Training and testing performance of lasso model

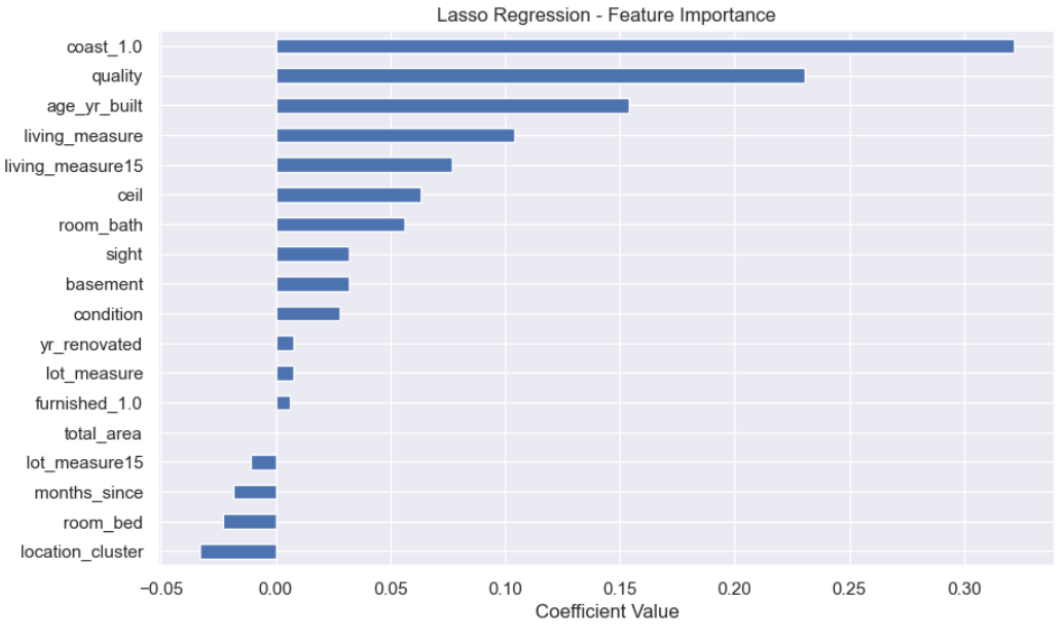


Figure 39: Feature importance of relevant features for the lasso model



Figure 40: Residual plot to understand the spread

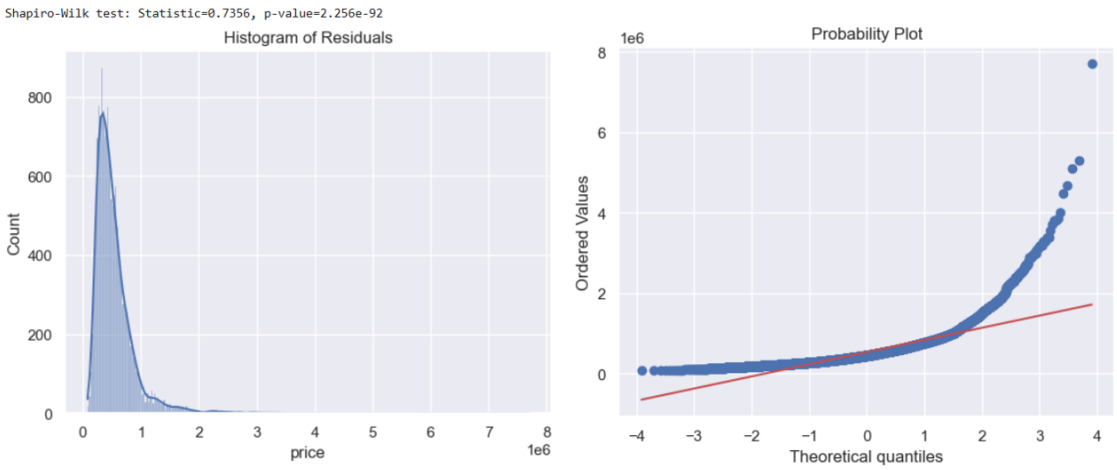


Figure 41: Residual distribution plot on left and Q-Q plot on right

Observation on assumptions

The residuals looks like having a pattern in the spread and the histogram of residuals is right skewed. The Shapiro-wilk test gives a test static of 0.736 with p-value <0.05, thus the null-hypothesis can be rejected. In other words, the residuals are not normally distributed and might require a preprocessing of the variable.

As the residual vs predicted plot has a funnel shape, it indicates heteroscedasticity.

From the features of the lasso model, it is clear that many high VIF features are not used in the model building.

Performance analysis

The r2 and adj r2 of training and testing data set are almost similar. The model is able to explain 70% variance of the dataset

- The RMSE and MAE are slightly higher for the testing dataset, but these values are almost in similar range - thus the data is not overfitting.
- The model is able to predict with mean error percentage of 24.5% which is slightly high

Model interpretation

The model can be interpreted based on the feature importance plot figure 38. The Lasso model tells us that the most expensive houses are typically **near the coast**, **higher quality**, **old built**, **larger living size**, **more ceils**, **more bathrooms** have **good number of sights**, **have basements**, **better condition**. Features like **number of bedrooms**, **lot size**, and **being on the market too long** and **location cluster** have a **negative impact** on price.

Model 3: Ridge Regressor

Ridge regression is a type of linear regression that is used when there are many features that may be correlated with each other. It helps prevent overfitting by adding a penalty to large coefficients in the model. This makes the predictions more stable and reliable, especially when working with complex or high-dimensional data. Ridge Regression is particularly useful in improving the accuracy of the model on new, unseen data.

Assumptions of Ridge Regression

Even though Ridge adds regularization, it still follows most **linear regression assumptions**:

1. Linearity
2. No perfect multicollinearity: Features shouldn't be perfectly duplicated. (Ridge can handle some correlation but not perfect duplication.)
3. Homoscedasticity
4. Normal distribution of residuals (errors)
5. Independence of errors

Let's look at the test and train metrics, residual analysis and observations on meeting the assumptions:

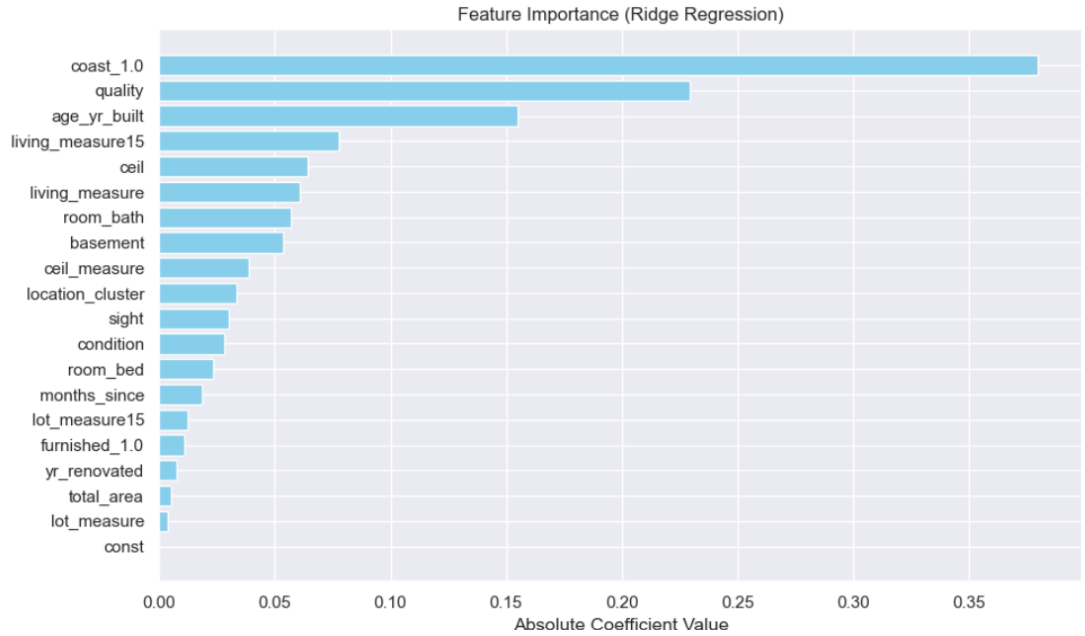


Figure 42: Feature importance in the descending order for relevant features of ridge regressor model

Training performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.307368	0.245233	0.651847	0.65138	1.886962

Testing performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.307835	0.244613	0.67488	0.67386	1.883509

Table 18:Performance metrics on training and test data set

Performance Analysis

- The Ridge Regression model achieved an R-squared of ~0.702 on training data and ~0.691 on test data, indicating it explains about 70% of the variance in house prices.
- The Root Mean Squared Error (RMSE) is approximately 192,378 for training and 219,048 for testing, which shows a reasonable level of prediction error given the scale of house prices.
- The Mean Absolute Error (MAE) also stays consistent across training and testing, around 121,924 and 128,977, showing stable performance.
- The Mean Absolute Percentage Error (MAPE) is around 24–25%, indicating that, on average, the predicted prices deviate from actual prices by about 1/4th of the true value

Residual Analysis:

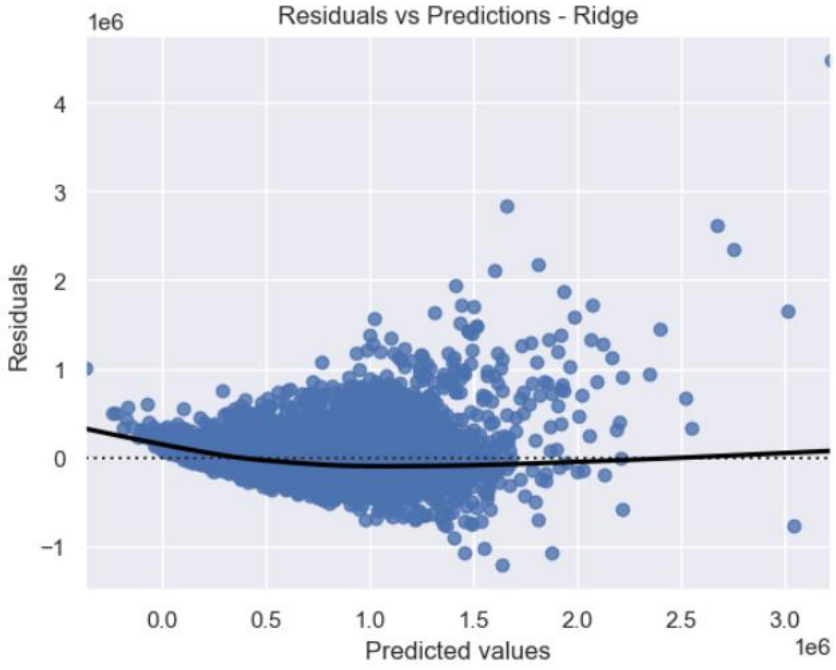


Figure 43: Residual plot for polynomial model

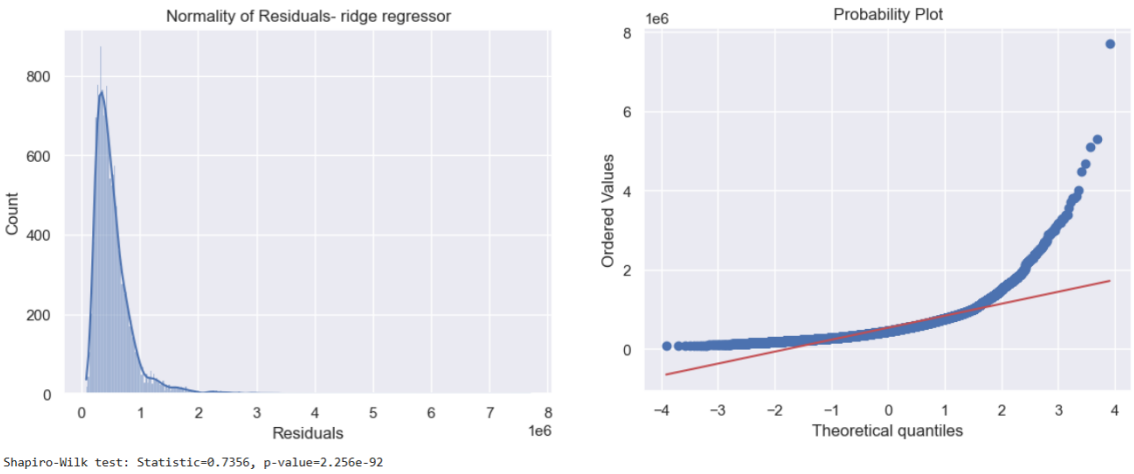


Figure 44: Residual distribution plot for ridge regressor and Q-Q plot

Model Assumptions and Diagnostics

- The Shapiro-Wilk test for residual normality failed (p-value < 0.05), suggesting the residuals are not normally distributed.[Statistic=0.736, p-value=2.25e-92]
- The Goldfeld-Quandt test for homoscedasticity also failed (p-value < 0.05), indicating the presence of heteroscedasticity (i.e., non-constant variance of residuals)[F-statistic: 1.1653607260503152, p-value: 2.0809045083955935e-11]

Residual vs predicted plot:

- It shows some patterns and outliers.
- Histogram shows right skewness and probability plot shows deviations at the right end tails- indicating outliers
- Although residuals show some pattern and outliers in visual checks, the model still performs well overall in predictive terms.

Model Interpretation

The model is interpreted based on the feature importance as shown in figure 41. The features importance is almost similar to that of the lasso or linear regressor models:

Positive coefficients: The biggest increase in house price comes from being on the coast, which adds a huge premium. Better quality and larger living spaces, including bigger ceilings, also raise the price a lot. Having more bathrooms also boost the price. Homes with basements, more ceiling measure and those that are furnished are worth more. Better condition adds value, as does having more bed rooms. Other parameters got smaller impact compared to the others.

Negative coefficients: None of the features have negative coefficients

Model 4: Polynomial Regression

Polynomial regression is an extension of linear regression that models the relationship between variables as a curved line instead of a straight one. It is useful when the data shows a non-linear pattern, allowing for better prediction accuracy by including squared or higher-power terms of the input variable.

Assumptions on polynomial regressor

- The relationship is linear in terms of model coefficients (even if the curve is non-linear in shape).
- Observations are independent of each other.
- Homoscedasticity of residuals
- No multicollinearity.
- Normality of residuals.

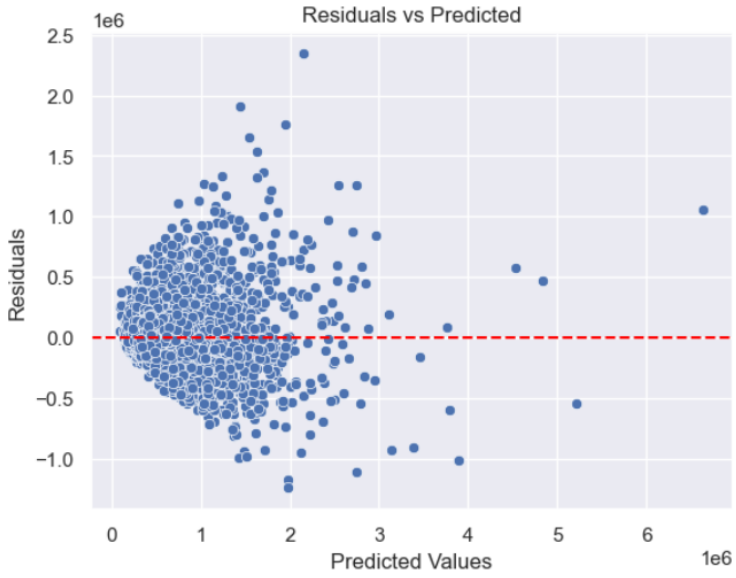


Figure 45: Residuals vs predicted values for polynomial regressor

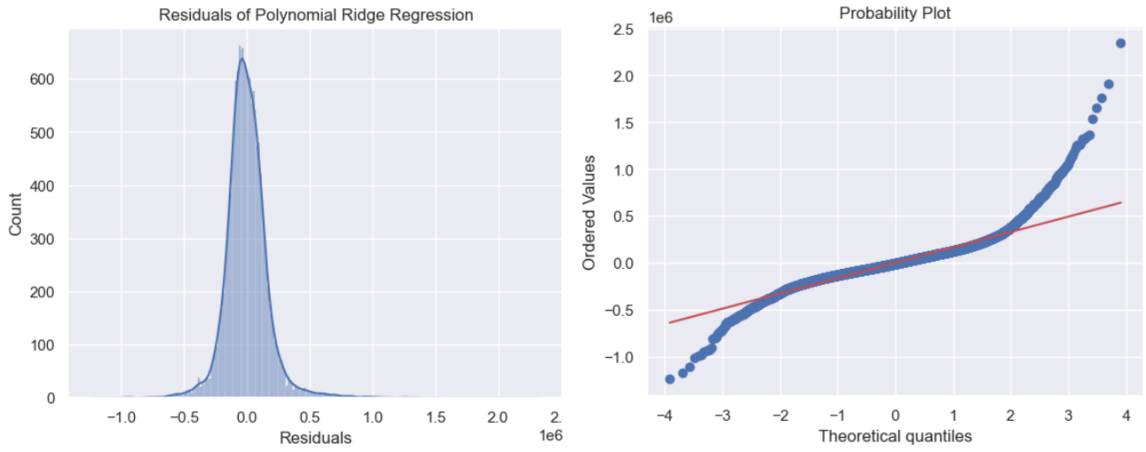


Figure 46:Residual distribution plot for polynomial regressor and Q-Q plot

```
Top Positive Predictors:
living_measure15 coast_1.0    105402.709143
living_measure coast_1.0      101142.884612
ceil_measure coast_1.0         92064.426337
coast_1.0 furnished_1.0        76924.416410
sight coast_1.0                64540.475043
const quality                  55143.800329
quality                        55143.800329
room_bath furnished_1.0        47160.346703
room_bath coast_1.0            44287.525156
age_yr_built                   39563.641397
dtype: float64
```

```
Top Negative Predictors:
basement furnished_1.0            -26827.800711
location_cluster age_yr_built     -29325.935187
ceil furnished_1.0                -31893.251316
age_yr_built furnished_1.0        -32601.083126
location_cluster furnished_1.0    -47313.651247
yr_renovated                      -56083.939828
const yr_renovated                -56083.939828
months_since coast_1.0            -56207.315935
age_yr_built coast_1.0            -94356.002056
quality coast_1.0                -132874.061724
dtype: float64
```

Table 19: Top 10 positive important features and top 10 negative important features

Model Performance Summary:

Training Performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	172641.277797	118277.339615	0.760016	0.759694	25.178747

Testing Performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	195256.166624	122868.793765	0.754771	0.754002	25.490109

Table 20: Training and testing performance of polynomial regressor

The Polynomial Regression model shows strong performance:

- Training R^2: 0.76 and Testing R^2: 0.75 — indicating good generalization. This is same value for Adjusted R^2.
- RMSE is reasonably low on both sets (172k train vs 195ktest).

MAPE ~25% shows slightly less accuracy in percentage terms.

Model Assumptions and Diagnostics

- Shapiro-Wilk Test (p-value ≈ 0): Residuals are not normally distributed.
- Goldfeld–Quandt Test (p-value = 1.13e-5): Suggests heteroscedasticity (non-constant variance).
- Q-Q Plot: Deviations at the tails — again pointing to non-normality.
- Histogram of residuals: Slightly right-skewed with long tails.
- Residuals vs Predicted Plot: Shows non-random patterns, especially at low predicted values — indicating model misfit or missing non-linearities.

Model interpretation

From the feature importance table, the most influential positive predictors include: **interaction** terms like (living_measure15, coast_1), (living_measure, coast_1), (ceil_measure , coast_1) and (coast_1, furnished_1)

The strongest negative predictors include: (quality coast1), (age_yr_built, coast_1), (months_since, coast1)

These indicate that interaction terms with coast_1.0 are quite dominant, and multicollinearity or overfitting could be a concern.

Model 5: Random Forest Regressor

Random Forest is a powerful machine learning model that builds many decision trees and combines their predictions to improve accuracy and reduce overfitting. It works well for both linear and complex, non-linear relationships. Each tree looks at a different random part of the data, and the final prediction is the average of all the trees' outputs.

To improve the model's performance, **RandomizedSearchCV** was used for **hyperparameter tuning**. This technique helps find the best combination of model settings (like number of trees, depth, etc.) by testing a random set of possibilities, making the model more accurate and efficient without excessive manual effort. Following table shows the best parameters obtained via the RandomizedSearchCV method.

```
Best parameters : {'n_estimators': 500, 'min_samples_split': 2, 'min_samples_leaf': 1, 'max_features': 'log2', 'max_depth': None}
```

Unlike the models built until now random forest regressor does not have any assumptions check.

Tuned Training Performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	58930.681743	34640.014789	0.972038	0.972	6.830778

Tuned Testing Performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	177023.829779	96901.511484	0.79843	0.797798	18.660363

Table 21: Training and testing data performance of random forest regressor

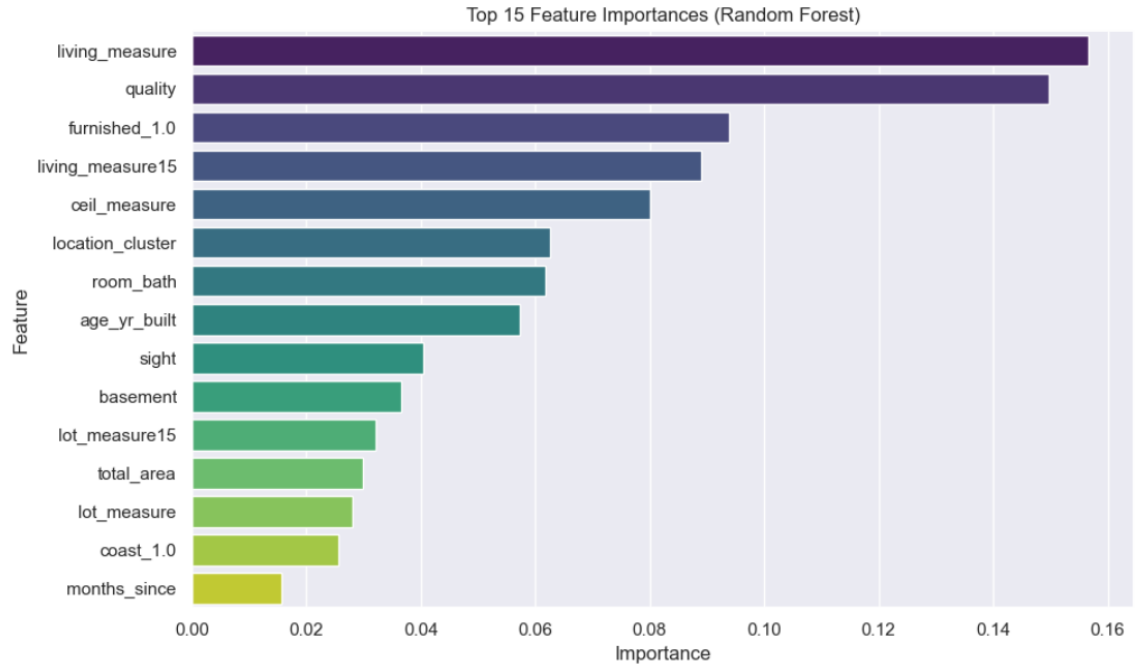


Figure 47: Feature importance of random forest regressor

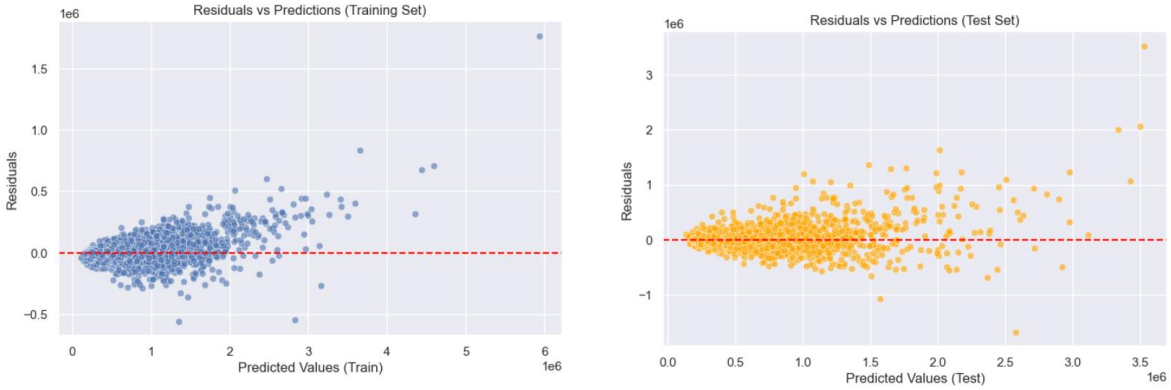


Figure 48: Residual plot for polynomial regressor. Left side is residuals for training set and right side plot is that of testing data

Model Performance:

The training performance is excellent, with an RMSE of ~58k and an R^2 of 0.972, indicating the model explains nearly 97% of the variance in the training set.

The testing performance shows some decline, with an RMSE of ~177k and R^2 of 0.798, which still represents a strong model but suggests overfitting**.**

The Mean Absolute Percentage Error (MAPE) is 6.8% for training and 18.6% for testing, indicating that the model is more accurate on seen data than unseen data – another indication of overfitting.

Residual Analysis

Residuals show some patterns at the left end of the plot and many outliers at the right end for both the training and testing data set.

- Train residuals are tightly clustered, model fits the training data well.
- Test residuals are also tightly clustered on left side of plot but more scattered on right side, slightly weaker generalization.
- Some heteroscedasticity is present, variance of residuals increases with the target value.

Model Interpretation

- Based on feature importance plot shown in figure 46, the most influential feature is **living_measure** contributing around 15% and **quality** (14.9%) to the model's predictions.
- Other important predictors include furnished_1.0 (9%), living_measure15(8.9%) and ceil_measure (7.9%)
- Location_cluster ,room_bath, age_yr_built also play a significant role.
- Less important features include yr_renovated, ceil, condition and room_bed

Model 6: SVM Regressor

Support Vector Regressor (SVR) is a machine learning model that tries to find the best line (or curve) that fits the data, while ignoring small errors within a certain margin. It is especially useful when the data has complex, non-linear patterns, and it can be adjusted to be more flexible using different kernel functions.

To improve model performance, **RandomizedSearchCV** was used for **hyperparameter tuning**. This helps automatically find the best settings (like kernel type, regularization strength, and margin width) to achieve higher accuracy and better generalization on unseen data.

Performance analysis

Training performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	362556.014331	218205.282016	-0.058384	-0.059804	42.177011

Tetsing performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	405349.71081	228859.729201	-0.056869	-0.060185	43.139737

Table 22: Performance metrics of SVM regressor on training and testing data

- The average RMSE is 362k which indicates high prediction error.
- The average R^2 score was -0.058, showing the model performed worse than a basic average prediction.
- The average MAPE was 40.5%, which is very high and reflects poor accuracy.
- Even after hyperparameter tuning using RandomizedSearchCV, the model's performance remained poor.
- **Based on these results, the SVR model is not suitable for this dataset.**

Residual Analysis

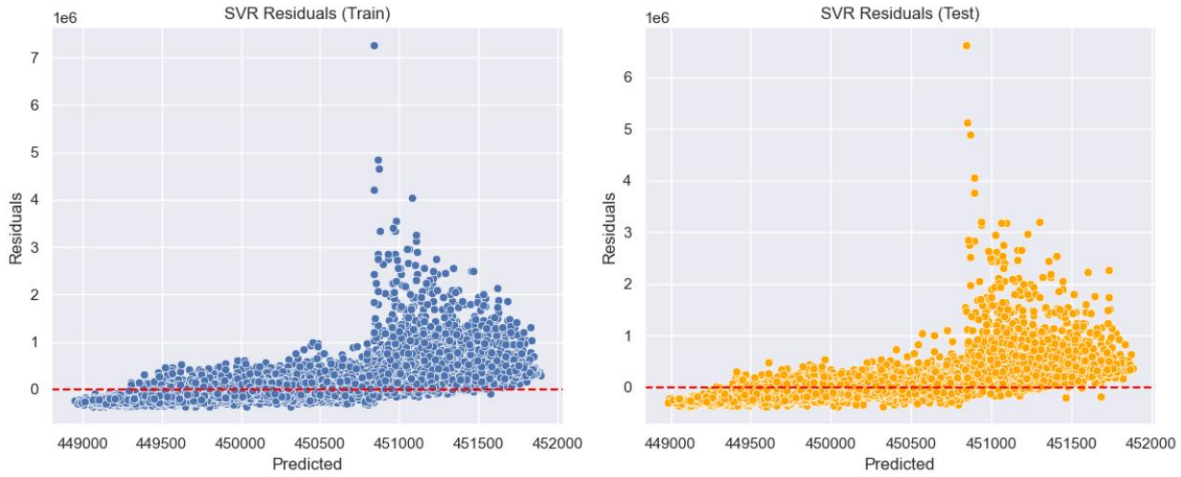


Figure 49:Residuals for SVM regressor. Left side plot is the residual for training data and right plot is for the testing data

Both the training and test data set shows similar residual plot. The residuals has some patterns in the both- the variance of residuals increases with increase in the price.

Feature importance and model interpretation

Please note that due to technical difficulties the feature importance plot for SVM is not available currently.

Model 7: XGBoost Regressor

XGBoost (Extreme Gradient Boosting) is a powerful machine learning algorithm that builds multiple decision trees one after another and combines their results to make better predictions. It is widely used for both classification and regression tasks because it's fast, handles missing values well, and can automatically capture complex patterns in the data.

In the context of regression, XGBoost tries to minimize prediction errors by boosting the performance of weak learners (simple trees) iteratively.

Here **GridSearchCV** method is a used to find the best combination of hyperparameters for the machine learning model. Common hyperparameters for tuning models like XGBoost include the number of trees (n_estimators), tree depth (max_depth), learning rate (learning_rate), and regularization terms like reg_alpha and reg_lambda.

Following are the best parameters obtained for the model:

```
Fitting 3 folds for each of 2916 candidates, totalling 8748 fits
Best Params for best_model_xgb6_1: {'xgb__colsample_bytree': 0.9, 'xgb__gamma': 0, 'xgb__learning_rate': 0.08, 'xgb__max_depth': 6, 'xgb__n_estimators':
210, 'xgb__reg_alpha': 0, 'xgb__reg_lambda': 2.0, 'xgb__subsample': 0.8}
```

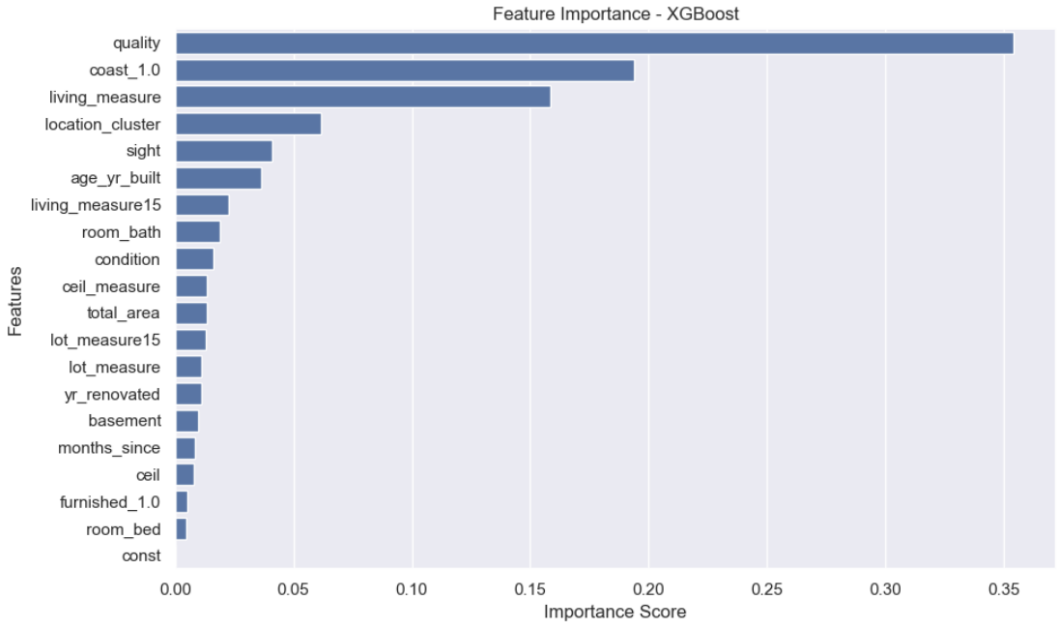


Figure 50: Feature importance in the descending order for the XGBoost regressor

Training performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	92112.384922	65160.894531	0.931683	0.931591	14.031901

Testing performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	165697.229814	91391.570312	0.823399	0.822845	17.291669

Table 23: Performance metrics on training and testing set for XGBoost regressor

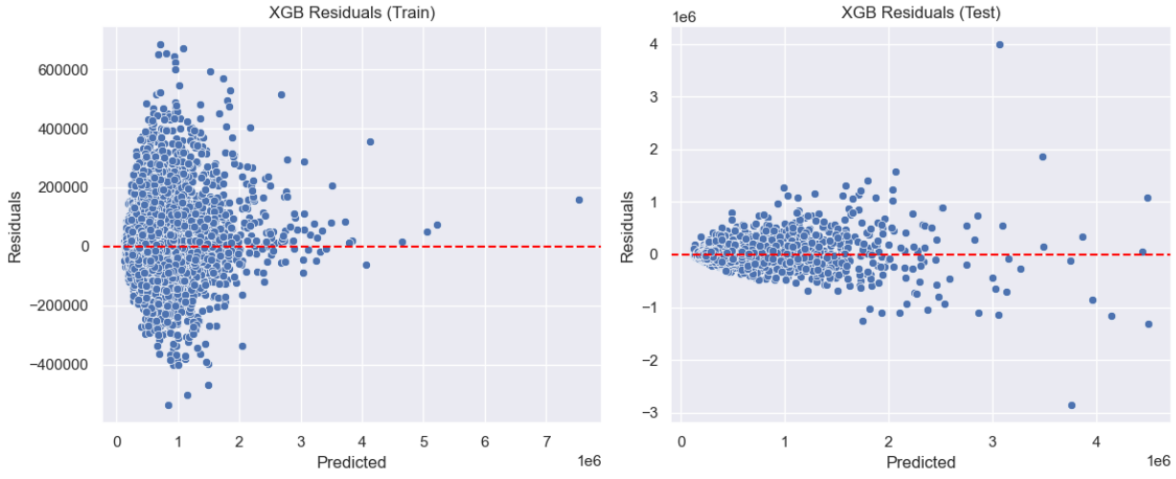


Figure 51: Residuals plot for training and testing set. Left plot is for training data residuals and right plot is for testing data residuals

Performance Analysis

- R squared and Adjusted R squared is 0.93 for training and 0.82 for testing
  This shows the model fits the training data very well but slightly less on new data, indicating **mild overfitting**
- RMSE is around ~92000 for training and ~165000 for testing
  The increase in RMSE on the test set confirms that the prediction errors are larger on unseen data
- MAE is about 65160 for training and 91391 for testing
  The difference is slightly large, but yet the average error is still reasonable
- MAPE is 14% for training and 17.2% for testing
  The model performs slightly worse on the test set, but the difference is not too big

Residual Analysis

The residual plot for the training set shows residuals mostly centred around zero but more concentrated toward lower predicted values, suggesting some underfitting for larger values. On the test set, residuals are slightly more spread out, especially for higher predictions, but still follow a relatively balanced pattern around the zero line, indicating the model generalizes reasonably well.

Model Interpretation

According to the feature importance scores, the most influential factor in predicting house price was overall **quality**, followed whether the house is located on the **coast** and by **living area size** . **Location** and **sight** also played notable roles.

On the other hand, variables like furnishing, ceil, months since house was on sale, basement has very less contribution in price prediction according to the model.

Model 8: AdaBoost regressor

AdaBoost Regressor is a type of ensemble learning method. It combines multiple weak learners (usually simple decision trees) to build a stronger predictive model. It does this by focusing more on the data points that previous models predicted poorly, and keeps improving in each round.

Using randomized seach cv, following are the best parameters:

```
Best Parameters: {'learning_rate': 0.1, 'loss': 'linear', 'n_estimators': 107}
```

Training perfromance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	222783.658348	166047.381316	0.600369	0.599832	39.393977

Testing performance:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	243457.730931	172093.442833	0.618751	0.617555	40.338561

Table 24: Performance metrics of AdaBoost regressor on training and testing data set

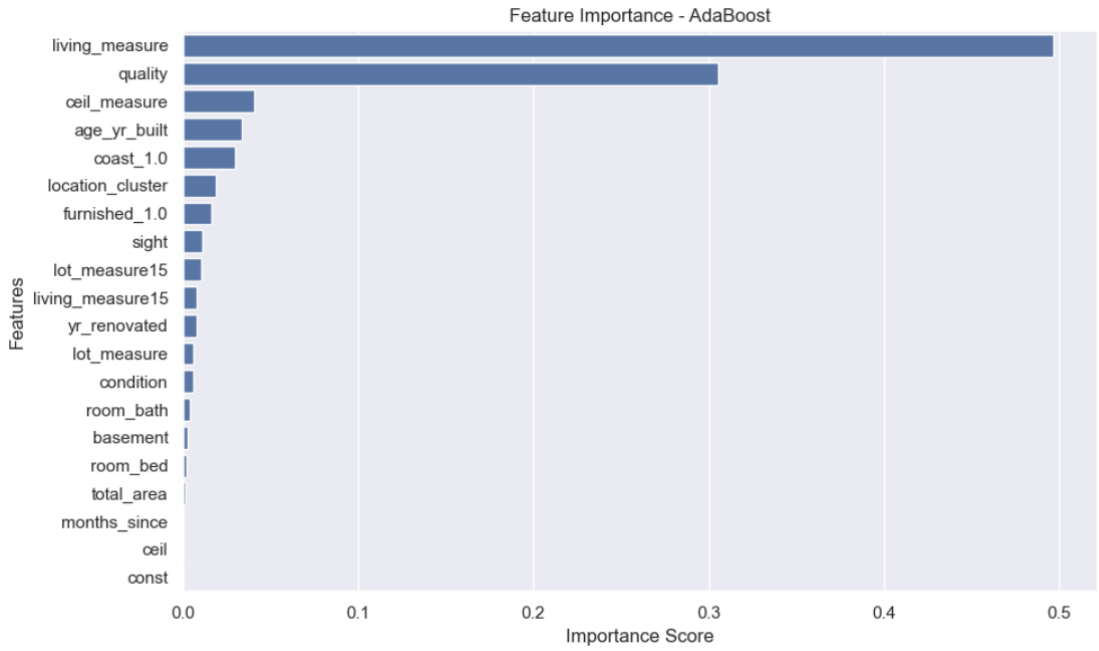


Figure 52: Feature importance for AdaBoost regressor in the descending order

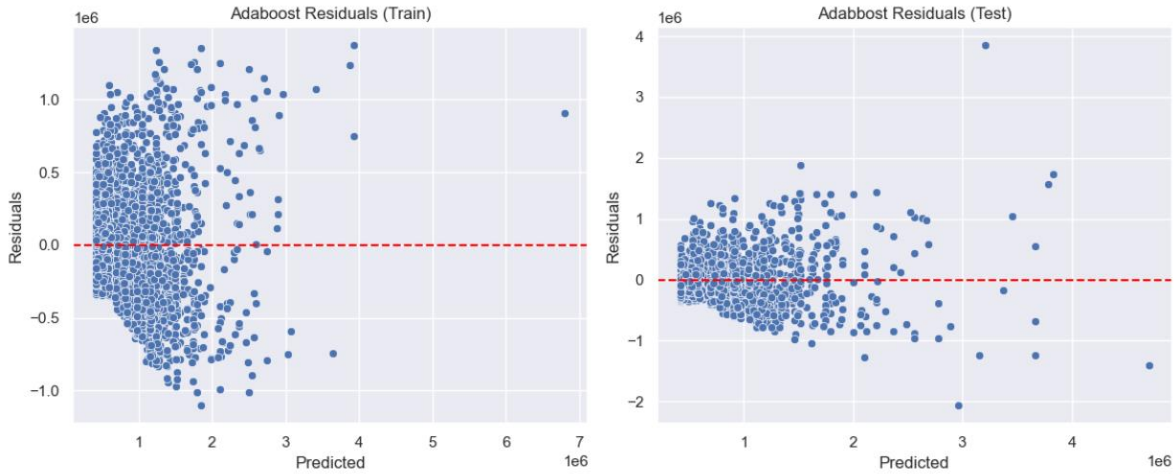


Figure 53: Residual plot for training and testing set

- The testing R-squared is 0.61 and training R-squared is 0.60. The testing R-squared is slightly higher than training, which indicates no overfitting and good generalization.
- The training RMSE is 222k and testing RMSE is 240k, showing stable error margins.
- MAPE is around 39 percent for training and 41 percent for testing, indicating similar prediction accuracy on both datasets, but a high amount of prediction error
- Overall, the model shows consistent performance, but with a high percentage of error.

Residual analysis

- The training residuals show a slight spread but are generally centered around zero.

- The test residuals are also centered around zero but have more variance, especially in higher predicted values.
- There is some pattern in the residuals, which suggests the model does not miss any obvious trend.
- Many outliers are present, but overall, the residuals are randomly scattered, indicating the errors are mostly unbiased.

Model Interpretation (Feature Importance)

- The most important feature is living_measure, contributing the most to the predictions.
- Other strong contributors are quality, latitude, and ceiling_measure.
- Furnishing status and coastal location also have a moderate influence.
- Features like room_bed, sight, lot_measure, and yr_renovated have relatively low impact.
- Variables like total_area, months_since, and ceiling contribute very little to the model.
- The intercept has zero importance

Final Model Selection: XGBoost Regressor

After evaluating multiple regression models including parametric and non-parametric ones, the **XGBoost Regressor** was selected as the most optimal model due to its high prediction accuracy and strong generalization ability on unseen test data.

- It achieved an R^2 of 0.93 on training data and 0.82 on test data, indicating excellent performance with relatively low overfitting.
- It has the low RMSE and MAE
- The MAPE is nearly 15-17 percent and not too high error percentage.
- The residual plot showed slight heteroscedasticity due to outliers, but still a good fit.

Model	Set	RMSE	MAE	Adj. R-squared	MAPE	R-squared
Xgboost regressor	Train	92112.384922	65160.894531	0.931591	14.031901	0.931683
	Test	165697.229814	91391.570312	0.822845	17.291669	0.823399
Random Forest Regressor	Train	58930.681743	34640.014789	0.972000	6.830778	0.972038
	Test	177023.829779	96901.511484	0.797798	18.660363	0.798430
Polynomial Regression	Train	0.286292	0.227469	0.697550	1.752284	0.697955
	Test	0.291108	0.229212	0.708340	1.765272	0.709252
Ridge Regression	Train	0.307368	0.245233	0.651380	1.886962	0.651847
	Test	0.307835	0.244613	0.673860	1.883509	0.674880
Linear Regression	Train	0.308436	0.246544	0.649093	1.897172	0.649422
	Test	0.309423	0.246251	0.670795	1.896439	0.671516
LASSO Regression	Train	0.307407	0.245293	0.651290	1.887402	0.651758
	Test	0.307407	0.245293	0.651290	1.887402	0.651758
ADABoost Regressor	Train	224519.537088	167932.292550	0.593572	40.085263	0.594117
	Test	224519.537088	167932.292550	0.593572	40.085263	0.594117
SVM Regressor	Train	362515.733601	218174.772933	-0.059569	42.173901	-0.058149
	Test	405311.720316	228831.603129	-0.059986	43.137263	-0.056671

Model Interpretation and Business Implications

The XGBoost model identified the following key drivers of house price:

- quality: Most influential feature. Better quality homes have significantly higher prices.
- coast_1.0: Houses near the coast are priced significantly higher.
- living_measure: Larger living area strongly contributes to higher price.
- location and sight: Certain locations and number of views also positively influence price.

These variables indicate that **quality of construction, coast, living measure, location and sight** are the top factors influencing price. These insights are critical for real estate decision-making. Following plot is visualizing the influence of these features on price.

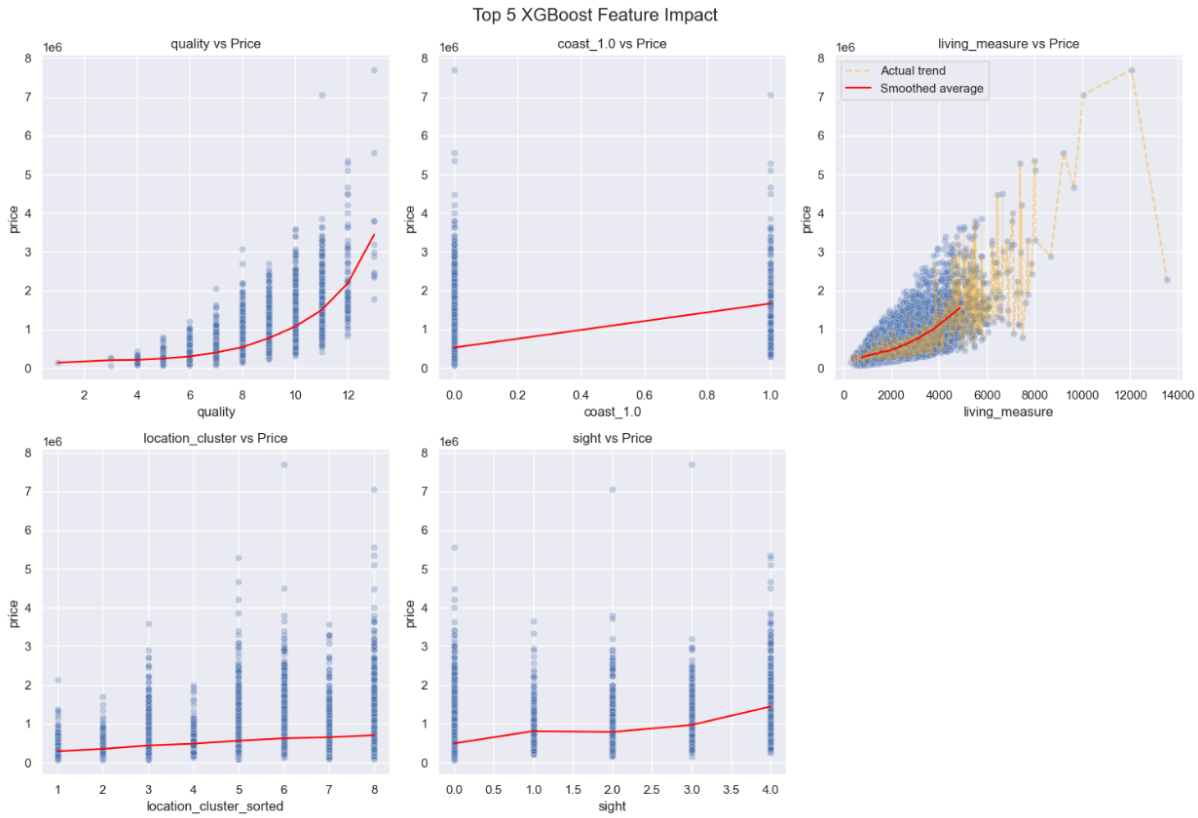


Figure 54: The scatterplot and lineplot of the top 5 features vs prices for XGBoost regressor. As predicted by the model, these plots show an positive relation between the target variable(price) and the explanatory variables. From top left to bottom following measures are plotted against price: quality, coast, living_measure, location and sight

Recommendations

For Real Estate Agents:

- Highlight construction quality in listings, especially for high-end buyers. Segment the listings based on quality ratings – Premium, Executive, Standard. Educate sellers on upgrades that improve the quality ratings
- Promote properties with large living areas and good coastal access.
- When valuing or pricing homes, be mindful of the location coordinates and number of sightings as they drive pricing substantially. Use the XGBoost model or polynomial model to price properties more accurately/ Also avoid overpricing large properties beyond the threshold where returns diminish. Similarly reassess the pricing strategy for coastal homes based on actual demand, quality and closeness to other amenities.
- Highlight spacious houses with floor plans and virtual tours. For small homes, emphasize smart layout and spaces.
- The model shows a positive relationship between the number of viewings (sight) and final price. Promote open houses more aggressively — via ads, digital platforms, and agents — to increase footfall and potentially bidding interest.
- Location cluster plays a major role in pricing. Organize open houses in high-demand localities more frequently or during weekends when traffic is higher.

For Buyers:

- Invest in better construction quality homes if long-term appreciation is the goal.
- Coastal or scenic properties may come at a premium but retain value better.
- Avoid overpaying for properties with low scores in key features like quality or living area.
- Larger homes costs more, but focus on layout efficiency also.
- Enquire about location-based price trends from the real estate agents to decide on the location